PROFESSIONAL SKILLS

Consulting, Discovery & Stakeholder Management

Discovery interviews, issue trees, stakeholder maps, SOWs, workshops, value cases and the hard conversations an embedded engineer has to run without authority.

consulting · discovery · stakeholder-management · mece · issue-trees · problem-framing · interviews · process-mapping · raci · governance · statement-of-work · scoping · workshops · prioritisation · fde

Study Pack · Comprehensive reference & interview prep · ~148 min read · 32,457 words

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Why an Engineer Needs Consulting Skills

A Forward Deployed Engineer is paid twice: once for the software, and once for the judgement about what software to build. The second payment is larger, harder to replace, and almost never taught. An engineer who can write a clean service but cannot run a discovery session will spend a career building the wrong thing beautifully. The consulting skill set — structured problem solving, disciplined questioning, stakeholder navigation, scoping, and honest communication — is what converts raw engineering capability into outcomes a customer will pay to repeat.

~70%
OF FAILED ENTERPRISE PROJECTS FAIL ON REQUIREMENTS AND STAKEHOLDERS, NOT CODE

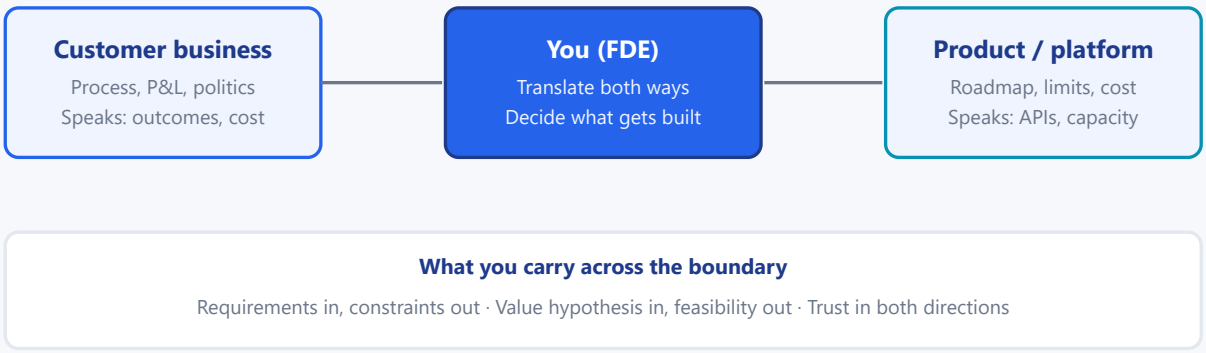
5-10x
COST MULTIPLIER OF FIXING A MISUNDERSTOOD REQUIREMENT AFTER BUILD VS DURING DISCOVERY

1 person
THE TYPICAL NUMBER OF PEOPLE WHO CAN ACTUALLY KILL YOUR PROJECT

<3 weeks
HOW LONG A SPONSOR WILL WAIT BEFORE DOUBTING AN ENGAGEMENT WITH NO VISIBLE OUTPUT

What the FDE role actually is

The title varies — forward deployed engineer, solutions engineer, delivery engineer, field engineer, embedded engineer — but the shape is constant. You are an engineer placed inside a customer's organisation, with their data, their constraints, their politics and their deadlines, and you are expected to produce working software plus the understanding that makes the software correct. You sit between three worlds and translate between them continuously.



The FDE sits on a boundary. Most of the job is deciding what crosses it, in which direction, and in what words.

The value an FDE adds beyond code

PROBLEM SELECTION

Choosing which of eleven candidate problems to solve first is worth more than any implementation choice. A mediocre solution to the highest-value problem beats an elegant solution to the fourth-ranked one, every time, in every engagement.

CONSTRAINT DISCOVERY

Nobody writes down that the warehouse team cannot use a phone on the floor, or that finance locks the ledger on the 3rd working day. These facts determine whether your design is usable. Only sustained presence surfaces them.

DECISION FORCING

Enterprises are decision-starved, not information-starved. An FDE who can take a fuzzy debate and turn it into "three options, here's my recommendation, I need an answer by Thursday" moves a project further than a week of coding.

CREDIBLE ESTIMATES

Because you have seen the data and talked to the users, you can say "four weeks" and be believed. Sales cannot do this. Remote architects cannot do this. It is a structural advantage of being embedded.

RISK SURFACING

Saying "the PDFs are scanned images, so OCR quality caps our accuracy around 85%" in week one is worth more than any feature. Risks found late become crises; risks found early become scope decisions.

TRANSFERABLE CAPABILITY

You leave. What remains is documentation, trained users, runbooks and a team that can extend the system. An FDE who leaves behind a system nobody can operate has failed, regardless of how well it works on the last day.

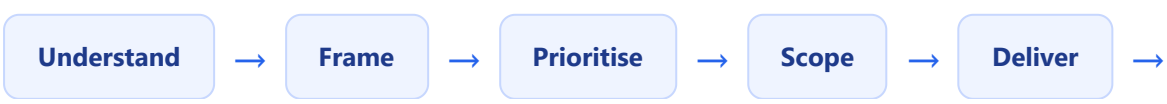
Engineer instincts that actively hurt you

Engineer instinct	Why it fails on a customer site	Replace with
"Give me the spec"	No spec exists. If one exists, it is wrong. The customer is describing a solution they imagined, not the problem they have.	Run discovery. Derive the spec and get it confirmed in writing.
"That's not technically interesting"	Value and technical interest are uncorrelated. The highest-ROI work is often a CSV import and three validation rules.	Rank by value, then find the interesting problem inside the valuable one.
"I'll build it properly first"	Three weeks of silence destroys sponsor confidence. Nobody can tell the difference between "building carefully" and "stuck".	Ship something visible every week, even if it is ugly.
"Just tell them it's impossible"	Blunt refusal ends the conversation and damages the relationship. Almost nothing is impossible; things are expensive or slow.	"Here's what that costs, here's the cheaper 80% version. Which do you want?"
Answering the question asked	The asked question is frequently the wrong one. "Can you add a Excel export?" often means "I don't trust your dashboard numbers."	Ask what they will do with the answer before answering.
Avoiding meetings to protect focus	The information that determines correctness only exists in people's heads and only emerges in conversation.	Treat 40% of your time as information gathering. It is engineering work.
Writing only for other engineers	The person who renews the contract reads two paragraphs and a number.	Write the executive version first, link the detail.

✦ **The single highest-leverage habit**

End every customer conversation by writing a short summary email within two hours: what you heard, what you decided, what you will do, what you need from them, by when. Do this for a year and you will out-perform engineers with twice your technical depth. It creates a written record, surfaces misunderstandings while they are cheap, and makes you the person whose version of events is the agreed one.

The consulting loop





Every section of this guide is a stage of that loop. The loop runs at multiple timescales simultaneously: once over a nine-month engagement, once per two-week sprint, and once inside a single forty-minute meeting. Learning to recognise which stage you are in — and refusing to skip stages because you are impatient to code — is most of the discipline.

A self-diagnostic

- ☐ Can you name the economic buyer of your current project, and the last thing they said to you directly?
- ☐ Can you state, in one sentence with a number in it, what the business gains if you succeed?
- ☐ Do you know who, specifically, would be annoyed if your project succeeded?
- ☐ Is there a written document, agreed by the customer, that lists what you will and will not deliver?
- ☐ If you disappeared tomorrow, could someone else operate what you have built from written material alone?
- ☐ Has your sponsor heard bad news from you before they heard it from anyone else?

Six "no" answers is normal for a strong engineer new to the role. Six "yes" answers is what separates an FDE from a contractor with a laptop.

Structured Problem Solving

Consulting firms teach structured problem solving because it is the only reliable defence against two failure modes: boiling the ocean, and jumping to the first plausible cause. The toolkit is small — decompose exhaustively, hypothesise early, test cheaply, cut ruthlessly — but it has to be practised until it is reflexive, because the pressure in a customer room always pushes toward unstructured flailing.

MECE: mutually exclusive, collectively exhaustive

A decomposition is MECE when every item belongs to exactly one branch and the branches together cover everything. It is a quality test for a list, not a philosophy. Non-MECE thinking is how you end up double-counting savings, missing an entire category of root cause, or arguing in circles because two people's categories overlap.

NOT MECE

"Why are orders delayed? Because of the warehouse, because of bad data, because of the night shift, because of SAP." Warehouse and night shift overlap; SAP and bad data overlap. You cannot size any branch without double-counting.

MECE

"Order delay = time to enter + time to validate + time to pick + time to ship." Four disjoint spans that sum to the whole. Each can be measured independently and the parts add to the total.

Useful MECE cuts you can reach for under pressure:

Cut	Branches	Use when
Process stages	Sequential steps that sum to total elapsed time	Anything with a cycle time or latency complaint
Algebraic	Profit = (price × volume) − (fixed + variable cost)	Financial outcomes; forces arithmetic honesty
Actor	Who does it: customer / agent / system / third party	Quality or error problems where blame is contested
Lifecycle	Acquire → onboard → use → renew → churn	Customer-facing or revenue problems
Internal vs external	Things we control / things we influence / things we endure	When the room is drifting toward helplessness
Volume / rate / mix	How many, how fast, what proportions	Capacity, throughput and cost-per-unit problems
People / process / technology / data	Four classic root-cause families	Generic fallback; weakest cut, use only when stuck

> ⚠ **The lazy cut**
>
> "People, process, technology" is the decomposition engineers reach for when they have not thought. It is barely MECE (a bad process is usually enforced by technology and tolerated by people) and it produces generic findings. Prefer a cut whose branches can be measured in the customer's own data: minutes, rupees, tickets, orders, defects.

The issue tree

An issue tree takes a single question and decomposes it, MECE, level by level, until the leaves are things you can actually measure or test in days. The goal is not completeness for its own sake; it is to make the whole problem space visible so that you can prune confidently and say to a customer "we are deliberately ignoring these four branches, and here is why."

Worked example. A B2B distributor's customer-service team is drowning. The stated problem: "our agents take too long to answer customer emails." The sponsor wants a chatbot. Before agreeing to build a chatbot, decompose.

TEXT

```
QUESTION: Why does it take 26 hours (median) to answer a customer email?

Total response time = QUEUE TIME + HANDLE TIME + APPROVAL TIME + SEND DELAY
                      (all four measured from the mail server + CRM timestamps)

1. QUEUE TIME  (email received → agent opens it)        median 19h  [73%]
   1.1 Arrival rate exceeds capacity
       1.1.1 Volume growing              → +14% YoY, measured
       1.1.2 Headcount flat              → confirmed, 9 FTE for 3 years
       1.1.3 Shift coverage gaps         → 0 coverage 18:00-08:00 = 14h of the 19h  *
   1.2 Routing failures
       1.2.1 Mail lands in shared inbox, no assignment rules → TRUE
       1.2.2 Misrouted to wrong region   → 6% of mails, re-queued once
   1.3 Triage cost
       1.3.1 Agent must open mail to know if it is urgent → no subject taxonomy
       1.3.2 No priority signal from CRM

2. HANDLE TIME (agent opens → drafts reply)              median 4h   [15%]
   2.1 Information gathering
       2.1.1 Agent must check 3 systems (CRM, SAP, shipping portal) → avg 11 min
       2.1.2 Credentials/session timeouts  → 2 min per lookup
   2.2 Composition
       2.2.1 No templates for the top 20 intents → TRUE, agents retype
       2.2.2 Free-text, high variance
   2.3 Interruption
       2.3.1 Agents also take phone calls → 40% of day

3. APPROVAL TIME (draft → approved)                      median 2h   [8%]
   3.1 Credit-note mails need supervisor sign-off → 11% of volume only
   3.2 Supervisor availability

4. SEND DELAY (approved → sent)                          median 1h   [4%]
   4.1 Batch send job runs hourly → trivially fixable

PRUNE: branches 3 and 4 together are 12% of total time. Park them.
FOCUS: 1.1.3 (shift gap) + 1.2.1 (no routing) = ~16h of the 26h median.
HYPOTHESIS: Auto-triage + assignment rules + off-hours auto-acknowledgement
            removes more time than any drafting assistant could.
TEST:       Pull 2 weeks of mail server logs. Compute response time with
            queue time artificially set to 1h. If median drops below 8h,
            the chatbot is solving the 15% problem.
```

✓ **What this buys you**

You walked in asked to build a chatbot. You walk out of week one with a measured tree showing that 73% of the delay is queueing, that a chatbot addresses at most 15%, and with a concrete recommendation that costs less and delivers more. That is the difference between an order-taker and a consultant. Crucially, you did not say "no" to the chatbot — you showed the arithmetic and let the sponsor choose.

Hypothesis-driven work

The slow way to solve a problem is to gather all the data, then analyse it, then conclude. The fast way is to form a specific, falsifiable hypothesis early and then design the cheapest possible test of it. Being wrong quickly is the mechanism; a hypothesis you cannot disprove is useless.

TEXT

```
HYPOTHESIS CARD

Hypothesis:   Off-hours arrival is the dominant driver of response latency.
Falsifiable?  Yes.
Prediction:   If true, ≥60% of median queue time is accumulated between
              18:00 and 08:00 local.
Disproof:     If off-hours accounts for <30%, the hypothesis is dead and
              we move to branch 1.2.
Cheapest test: Export 14 days of message headers (received_at, first_open_at)
              from the mail server. 1 CSV, ~2 hours of work. No access
              requests needed beyond the mail admin.
Owner:        Me
By:           Thu 14:00
If confirmed: Price up (a) 24/5 follow-the-sun coverage via the Manila team,
              (b) auto-acknowledge + auto-triage bot.
If refuted:   Test routing hypothesis (1.2.1) using CRM assignment logs.
```

Rules that make this work in practice:

1. **Write the hypothesis before you look at the data.**
   Otherwise you will find a pattern in anything and call it a finding. Writing it down also lets you be publicly wrong, which builds far more credibility than quiet rightness.

2. **State the number that would disprove you.**
   "A lot" and "most" are not predictions. "At least 60%" is.

3. **Choose the test that is cheapest, not the one that is most rigorous.**

A two-hour CSV analysis that moves you from 50% to 85% confidence is worth more than a two-week study that reaches 95%. You are making a decision, not publishing.

4 **Timebox the test before you start it.**
"If I cannot get this data by Thursday, I will go and watch three agents work for an hour instead." Data access in enterprises fails often; always have the observational fallback.

5 **Kill hypotheses out loud.**
Telling the sponsor "I thought X, I tested it, I was wrong, here's what I now think" is one of the strongest trust-building moves available.

The 80/20 cut

Pareto is not a slogan, it is an instruction to go and count. In almost every enterprise dataset the distribution is brutally skewed, and the skew is where your scope lives.

TEXT

```
THE COUNT THAT SETTLES THE ARGUMENT

  Intent / category          Volume   Cum %    Avg handle   Total minutes  Cum %
  ──────────────────────────────────────────────────────────────────────────────
  "Where is my order?"        4,210    31%        6 min        25,260       21%
  "Invoice copy request"      2,880    52%        4 min        11,520       31%
  "Price / availability"      1,950    67%        9 min        17,550       45%
  "Return authorisation"      1,120    75%       22 min        24,640       65%
  "Change delivery address"     690    80%        7 min         4,830       69%
   ... 94 other intents       2,700   100%       13 min        36,180      100%

READ IT TWICE:
 - By VOLUME, the top 5 intents are 80% of mails.
 - By MINUTES, returns (1,120 mails) cost almost as much as order status
   (4,210 mails). A per-ticket-volume view would have under-weighted returns.

SCOPE DECISION: automate intents 1 and 2 (templated, deterministic, 36% of
minutes, low risk). Do NOT automate returns yet - high handle time means
high judgement content and high error cost. Revisit in phase 2.
```

⚠ **Always Pareto on the unit that matters**
Teams routinely Pareto on ticket count and then discover their automation saved the cheapest tickets. Pareto on cost, minutes, revenue at risk, or customer pain — whichever is the thing the sponsor actually cares about. If you are unsure which, ask: "if we halved this category, what would you notice?"

Day-one answer and the hypothesis pyramid

Consultants write the answer on day one, knowing it is wrong, because an explicit wrong answer organises all subsequent work. The "day-one answer" is a one-page straw man: what we think the problem is, what we think the solution is, what the value is, and what we would have to believe for that to be true. Every piece of fieldwork then either supports or breaks a specific belief.

TEXT

```
DAY-ONE ANSWER (straw man) - Distributor CS engagement - v0.1 - Mon

SITUATION  CS team of 9 handles ~13,500 customer emails/month. Median
           response 26h vs a 4h commitment in their customer charter.
           Two key accounts have cited responsiveness in QBRs.

COMPLICATION Volume +14% YoY with flat headcount. Hiring is frozen until Q3.

ANSWER     We believe response time is a QUEUEING problem, not a DRAFTING
           problem, and that auto-triage plus off-hours acknowledgement
           cuts median response from 26h to under 6h within 8 weeks,
           without new headcount.

WHAT WE MUST BELIEVE FOR THAT TO BE TRUE
  B1  Most queue time is accumulated outside business hours.      [test: mail logs]
  B2  Intent can be classified from subject+body at >90% accuracy. [test: label 300]
  B3  Customers accept an automated acknowledgement.              [test: ask 2 account
  B4  The mail platform permits API access for assignment.        [test: ask IT]
  B5  Agents will trust and use an assigned queue.                [test: shadow 3 agen

KILL CRITERIA  If B1 is false AND B2 is false, this answer is wrong and we
               pivot to the drafting-assistant branch.

NEXT 5 DAYS    B1 Tue, B4 Tue (long lead), B2 Thu, B5 Wed, B3 Fri.
```

Say "straw man" out loud

When you present a day-one answer, label it: "this is a straw man, I expect to be wrong in at least two places, and I would like you to attack it." People engage far more honestly with something presented as provisional, and you get the corrections in minute three instead of week six.

Common structuring failures and their tells

Failure	Tell	Fix
Boiling the ocean	Your workplan has 14 workstreams and no sequencing	Force-rank, pick two, write down what you are deliberately not doing
Solution-first	The technology was chosen before the problem was measured	Write the issue tree anyway; it either validates the choice or saves the client
Analysis paralysis	Week four, no recommendation, "still gathering data"	Force a day-one answer and a kill criterion; set a decision date
Anecdote as evidence	"The ops manager said it's all the night shift"	Treat it as a hypothesis, not a finding. Go count.
Non-MECE sizing	Your savings categories sum to more than 100% of the cost base	Redo the decomposition algebraically so branches are disjoint
Measuring what is easy	Rich analysis of a 4% branch	Size every branch roughly before analysing any branch deeply
One-level trees	Leaves are still abstract ("bad data quality")	Keep splitting until a leaf names a specific field, system or step

Drill: build a tree in ten minutes

1. **Write the question as a single sentence with a number in it.**
   "Why is our median response time 26 hours when our charter promises 4?" Not "why is CS slow?"

2. **Pick a cut and justify it.**
   Prefer additive cuts where branches sum to the measured whole.

3. **Split to three levels maximum.**
   Depth beyond three is usually procrastination.

4. **Size every branch, even roughly.**
   Guess if you have to, and mark guesses with a `?`. An order-of-magnitude guess is enough to prune.

5. **Mark the two branches you will pursue and strike through the rest.**
   Show the struck-through branches to the customer; it proves you considered them.

6 **Write one hypothesis card per pursued branch.**

Framing the Problem

More engagements are destroyed by a badly framed problem than by any technical failure. A bad frame is invisible: everyone nods, work begins, and six weeks later two people discover they were solving different things. Framing is the discipline of writing the problem down so precisely that disagreement becomes impossible to hide, then pressure-testing the statement until it earns the work.

The problem statement canvas

Seven fields. If you cannot fill all seven, you do not yet understand the problem well enough to scope it — and that gap is itself the most useful finding of week one.

TEXT

```
PROBLEM STATEMENT CANVAS

1. WHO          Who has this problem, specifically? Named role, named team,
                how many of them.
2. WHAT         What is happening that should not be, or not happening that
                should? Observable behaviour, not interpretation.
3. SIZE         How big is it, in a unit the business already tracks?
                Money, hours, tickets, defects, delay, risk exposure.
4. SINCE/WHEN   When does it occur? Always, month-end, peak season, since a
                system change? Frequency and trigger.
5. EVIDENCE     How do we know? Which system, which report, which sample.
                If the only evidence is someone's opinion, say so.
6. IMPACT       What is the consequence of not fixing it? Who feels it, and
                what happens over the next 12 months if nothing changes.
7. CONSTRAINTS  What is off the table? Budget cap, no new headcount, must
                run on-prem, cannot touch SAP, regulatory deadline.

SUCCESS         What is true at the end that is not true now, measured how,
                by when, verified by whom.
```

Worked examples: bad versus good

BAD

"We need to improve our data quality."

WHY IT FAILS

No actor, no size, no evidence, no constraint, and no definition of improved. It licenses infinite work and permits no acceptance test. Every stakeholder will fill the blanks with their own project.

TEXT

```
GOOD (same underlying issue, properly framed)

WHO     The 14 pricing analysts in EMEA Commercial Operations.
WHAT    They manually correct customer master records before every quarterly
        price review because country codes, VAT ids and parent-company links
        are inconsistent between SAP and Salesforce.
SIZE    ~1,900 records corrected per quarter, measured from the change log.
        At 7 min/record that is 222 hours per quarter = 0.55 FTE.
        Separately, 3 mispriced contracts in FY24 traced to stale parent
        links; combined credit notes EUR 184,000.
WHEN    Every quarter, concentrated in the 9 working days before the review.
EVIDENCE SAP CDHDR change log export (Q1–Q4 FY24); Salesforce field history;
        interview notes with 4 analysts; the 3 credit-note cases from finance.
IMPACT  Analysts work weekends in review weeks; two resigned citing it.
        Pricing accuracy risk is now on the CFO's control register.
CONSTRAINTS No change to SAP master data governance before the S/4 migration
        (Oct). No new headcount. Must not require analysts to learn SQL.

SUCCESS By 30 June, manual corrections per quarter fall below 400, verified
        from the same SAP change log, with no increase in mispriced contracts.
        Reviewed by the Head of Commercial Ops.
```

A second pair, from a different domain:

BAD

"The warehouse system is slow and the team hates it."

GOOD

"Pickers at the Bhiwandi DC wait a median of 11 seconds after each scan before the next item appears, on a handheld that scans ~340 times per shift. That is ~62 minutes of standing per picker per shift across 22 pickers. Measured by screen recording on 6 shifts. Peak-season overtime is budgeted at INR 41 lakh largely to absorb this. Constraint: handhelds cannot be replaced this financial year."

> **The number does most of the work**
> Notice that in both good statements, the moment a number appears the conversation changes character. "Data quality is bad" invites opinion. "222 hours per quarter, from the change log" invites verification and then action. If you can only improve a problem statement in one way, add a defensible number and say where it came from.

The "so what?" test

After every statement of fact, ask "so what?" until you reach something a budget-holder cares about. If you cannot get there in three steps, the finding is not worth presenting.

TEXT

```
FACT   "38% of inbound invoices arrive as scanned PDFs rather than XML."
SO WHAT? "Those require manual key-in by the AP team."
SO WHAT? "AP spends 1,400 hours/year keying and correcting them, and the
          error rate on keyed invoices is 2.3% vs 0.1% on XML."
SO WHAT? "That is 0.7 FTE plus roughly EUR 90k/yr of duplicate-payment
          recovery work, and it is why month-end close slips two days."
          ⟵ STOP. Month-end close slipping is a CFO-level pain.

Compare a chain that dies:
FACT   "The ingestion service is written in Python 2."
SO WHAT? "It is end-of-life and we cannot patch it."
SO WHAT? "There is a security risk."
SO WHAT? "…we should upgrade it."
          ⟵ Never reached a business consequence. Either find the real
              consequence (audit finding? vendor support cliff? blocked
              feature?) or stop presenting it as a headline.
```

Symptom, cause and the Five Whys done properly

The Five Whys is usually performed badly: a single chain of plausible-sounding assertions, none verified, terminating wherever the facilitator wanted to arrive. Done rigorously it has three rules — branch where there are multiple causes, verify each step with evidence before descending, and stop at the first cause you can actually act on rather than drilling to philosophy.

TEXT

```
FIVE WHYS — RIGOROUS FORM
Incident: 1,100 customers received a duplicate shipping notification on 3 Feb.

WHY 1  Why did customers get two notifications?
       → The notification job ran twice for the same batch.
       EVIDENCE: job scheduler log shows two executions, 02:14 and 02:51.

WHY 2  Why did it run twice?
       → The 02:14 run appeared to fail and an operator re-ran it manually.
       EVIDENCE: ops chat transcript 02:46–02:50; manual trigger in audit log.

WHY 3a Why did it appear to fail?            [BRANCH A — detection]
       → The job exited non-zero after sending, during the cleanup step.
       EVIDENCE: exit code 1, stack trace in /var/log/notify/2024-02-03.log
       shows failure in temp-file deletion AFTER the send loop completed.

WHY 3b Why did a manual re-run duplicate?    [BRANCH B — idempotency]
       → The job has no idempotency key and no record of what it sent.
       EVIDENCE: code review, notify_batch.py has no sent-ledger write.

WHY 4a Why did a cleanup failure exit non-zero for the whole job?
       → Cleanup is inside the same try block as the send; any exception
          fails the job regardless of stage.
WHY 4b Why was there no sent-ledger?
       → Written in 2019 as a one-off; never hardened. No design review.

WHY 5  Why was a one-off script running a customer-facing nightly process?
       → No inventory of production jobs; no ownership assigned.
       EVIDENCE: no entry in the service catalogue; no on-call owner.

ACTIONABLE STOPPING POINTS (do not go further — "why is there no process
culture?" is not actionable this quarter):
  A. Separate cleanup from send; cleanup failure must not fail the job.   [1 day]
  B. Add a sent-ledger keyed on (batch_id, customer_id); skip duplicates. [3 days]
  C. Runbook entry: never re-run notify jobs manually; escalation path.   [2 hours]
  D. Add notify_batch to the service catalogue with a named owner.        [1 hour]

NOTE A and B are independent fixes to independent causes. A single-chain
Five Whys would have found only one of them.
```

⚠ **Two abuses to refuse**

First, the chain that terminates at "human error" — that is where analysis stops being useful, because people will always eventually err; ask why the system permitted the error to have that consequence. Second, the chain that terminates at "we need more budget/headcount" on the first why, which is politics wearing analysis as a costume. In both cases say: "that may be true, but it is not actionable at our level. What is the nearest cause we can change?"

Reframing moves

When a problem statement feels stuck, these six moves usually unstick it. Each is a question you can ask out loud in a meeting.

Move	Question to ask	Example shift
Widen	"What is this a symptom of?"	"Reduce report build time" → "Why does anyone need this report weekly?"
Narrow	"Which 10% of cases cause most of the pain?"	"Automate claims" → "Automate motor claims under INR 25,000"
Invert	"What would we do to make this much worse?"	Exposes the levers: more handoffs, more approvals, less visibility
Change the actor	"Who else could do this step?"	"Agents are slow" → "Why is the customer not doing this themselves?"
Change the unit	"Per what? Per order, per hour, per rupee?"	Cost per ticket hides that 3% of tickets are 40% of cost
Delete the step	"What if this activity simply did not happen?"	The best automation is elimination; ask before you optimise

✦ **The question that saves projects**

"If we do nothing at all, what happens?" Ask it in the first week, with a straight face. Sometimes the honest answer is "nothing much", and you have just saved everyone six months. More often the answer is specific and urgent — an audit, a contract renewal, a regulatory date, a competitor — and now you know the real driver, which is almost never what the kickoff deck said.

Getting the frame agreed in writing

A frame that lives only in your head is not a frame. Send this after discovery week one, and do not start building until someone replies.

TEXT

Subject: Problem statement for sign-off – CS response time – please confirm by Thu

Hi Priya,

Before we commit engineering time I want to make sure we are solving the same problem. Here is what I believe we agreed; please reply with corrections or "confirmed" by Thursday and I will treat it as our baseline.

PROBLEM
Customers emailing customer-service wait a median of 26 hours for a first substantive response, against the 4-hour commitment in the service charter. Source: 14 days of mail-server timestamps joined to CRM first-touch events, n = 6,310 messages.

WHY IT MATTERS
Two key accounts (Rheinmetall, Tata Steel) raised responsiveness in their Q4 QBRs. Renewal value at those two accounts is EUR 2.1m, due in September.

WHAT WE ARE SOLVING
Queue time – the gap between a message arriving and an agent opening it – which is 19 of the 26 hours.

WHAT WE ARE EXPLICITLY NOT SOLVING (this phase)
 – Draft quality or tone of agent replies
 – The credit-note approval workflow (8% of elapsed time)
 – Phone channel
 – Anything requiring changes to SAP

CONSTRAINTS I HAVE ASSUMED – correct me if wrong
 – No new headcount before Q3
 – Mail platform is Exchange Online; IT will grant Graph API read by 12 Feb
 – Any automated customer-facing message needs Legal sign-off (2 weeks)

SUCCESS
Median first-response time below 6 hours, measured on the same mail-server join, sustained over 4 consecutive weeks, by 30 April.

If any of this is wrong, the cheapest moment to tell me is today.

Thanks,
Arun

Discovery Interviews

Discovery interviewing is the highest-yield activity available to an FDE and the one engineers do worst. The failure mode is predictable: you arrive with a solution in mind, ask questions that confirm it, write down conclusions instead of words, and leave with a validated misunderstanding. Done well, an hour with the right person collapses weeks of speculation — but only if you treat it as fieldwork with a protocol, not as a chat.

Before the interview

1. **Know what you are trying to learn.**
   Write the two or three things you must leave knowing. If you cannot write them, cancel and think first.

2. **Research the person.**
   Role, tenure, what they are measured on, who they report to, what they said in any prior meeting. Five minutes on their LinkedIn and the org chart prevents the insult of asking what they do.

3. **Send a short pre-read.**
   Two paragraphs: why you want to talk, what you will ask about, how long, and that there is no preparation needed. Reduces defensiveness enormously.

4. **Choose your opening and your closing.**
   The first ninety seconds set whether this is an interrogation or a collaboration. The last two minutes are where you get referrals and artefacts.

5. **Arrange note-taking.**
   Ideally a second person writes while you talk. If alone, ask permission to type, or record with explicit consent — and never record without it.

6. **Budget silence.**
   Plan 40 minutes of questions for a 60-minute slot. The gold arrives in the gaps.

TEXT

```
PRE-READ EMAIL (send 2 days ahead)

Subject: 45 min on how order exceptions actually work — no prep needed

Hi Martin,

I'm Arun, the engineer working with Priya's team on customer-service
response times for the next few months. Before we design anything I want
to understand how the work really happens, and I'm told you've handled
order exceptions longer than anyone.

I'd like 45 minutes to ask you about:
 - how an exception reaches you and what you do with it
 - where you have to go to get information
 - what makes a bad day bad

There's nothing to prepare — I want your description, not a polished one.
If you can, having your actual queue open during the call would help; I'd
rather look at real examples than talk in the abstract.

I'll send you my notes afterwards so you can correct anything I got wrong.

Thanks,
Arun
```

"I'll send you my notes so you can correct them"

Promise this at the start of every interview and then do it. It transforms the dynamic: you are no longer extracting information to be used against them, you are co-authoring a record. People relax, correct themselves, and volunteer more. It also gives you a written, stakeholder-validated artefact for free.

Open, closed, and the laddering technique

Type	Form	Use for	Danger
Open	"Walk me through the last time you..."	Discovery, surfacing the unexpected	Rambling; needs steering
Closed	"Does the system assign it, or do you pick?"	Confirming, pinning down specifics	Leads the witness; use late
Laddering down	"What does 'difficult' mean concretely?"	Converting abstraction to fact	Can feel like interrogation; soften
Laddering up	"Why does that matter to you?"	Finding motive and real priorities	Three levels maximum before it feels like therapy
Contrast	"How is that different from a normal day?"	Isolating variables	None; underused
Projective	"If you had a new colleague, what would you warn them about?"	Getting criticism people will not state directly	None; very effective

TEXT

```
LADDERING IN ACTION (verbatim from a real-shaped interview)

Me:      Walk me through this morning.
Martin:  Usual. Came in, cleared the overnight mess, then exceptions.
Me:      "Overnight mess" - what's in that?          [ladder DOWN]
Martin:  Orders that failed the credit check while nobody was here.
Me:      How many, typically?                        [quantify]
Martin:  Twenty, thirty. Monday it's sixty.
Me:      What do you do with each one?               [ladder DOWN]
Martin:  Look it up in SAP, see if the customer's over limit or if it's
         just the old address thing again.
Me:      "The old address thing"?                    [ladder DOWN - gold]
Martin:  If the ship-to doesn't match what's in the credit system it fails
         even though the customer's fine. Probably half of them.
Me:      Half of sixty, three times a week. What do you do then?
Martin:  Override it. Takes a minute each.
Me:      And why does that matter - what happens if they sit there?
                                                      [ladder UP]
Martin:  They don't get picked. If it's after 11 they miss the truck and
         it's next day. Customer rings up. Then it's my afternoon gone.

FINDING: ~50% of overnight credit-check failures are a false positive from
a ship-to address mismatch. ~15 per weekday, 30 on Monday. Each causes a
day's delivery slip if not cleared by 11:00. Nobody had mentioned this in
any workshop because to Martin it is just Tuesday.
```

> **Where the value was**
>
> Nobody asked for this to be fixed. It did not appear on any requirements list. It is a deterministic data-matching rule worth roughly 3,900 delayed orders a year, and it surfaced because one follow-up question was asked about a throwaway phrase. Chase every piece of jargon and every casual euphemism — "the old address thing", "the usual Friday problem", "the spreadsheet" — these phrases are where undocumented process lives.

The question bank

Not a script to be read in order. Pick six to ten per interview according to the goal, and let the conversation go where it wants.

OPENING AND FRAMING (BUILD SAFETY FIRST)

TEXT

```
1.  Tell me what you're responsible for – in your words, not your job title.
2.  How long have you been doing this, and what did you do before?
3.  What does a good day look like? What does a bad day look like?
4.  What are you measured on? What does your boss ask you about?
5.  If I shadowed you for a week, what would surprise me?
6.  Who else should I be talking to, and what will they tell me that you won't?
```

UNDERSTANDING THE CURRENT PROCESS

TEXT

```
7.  Walk me through the last one you handled – start to finish, out loud.
8.  How does work arrive at you? How do you know there's something to do?
9.  What do you do first? And then? [keep saying "and then?"]
10. Where do you have to go to find information to do this?
11. How many systems do you touch in a typical case? Can you show me?
12. What do you do when the system doesn't let you do what you need to?
13. Is there a spreadsheet involved? [There is always a spreadsheet.]
14. What happens to it after you're done? Who gets it next?
15. How do you know whether you did it right?
16. What's the rule you follow that isn't written down anywhere?
```

VOLUME, VARIATION AND EXCEPTIONS

TEXT

```
17. How many of these in a day? A week? Is that steady, or peaky?
18. When is your busiest period? What's different about it?
19. What proportion go straight through without any intervention?
20. Describe the weird ones. What makes a case hard?
21. What's the longest one you've ever dealt with, and why was it long?
22. When was the last time something went properly wrong? What happened?
23. Which customers / products / regions behave differently to the rest?
24. What do you do when you're not sure? Who do you ask?
```

PAIN, COST AND WORKAROUNDS

TEXT

```
25. What's the most frustrating part of this job?
26. What do you spend time on that feels pointless?
27. What do you do twice because of how the systems work?
28. If you could delete one step entirely, which one?
29. What do you do manually that you think a computer should do?
30. What have you built yourself to cope? [macros, scripts, templates, lists]
31. How much of your week goes on this? Honestly - not the official number.
32. What happens if this doesn't get done? Who notices, and when?
```

DATA AND SYSTEMS

TEXT

```
33. Where does this data come from originally? Who types it in first?
34. Which fields do you actually trust? Which do you ignore?
35. What's the field everyone fills in wrong?
36. How do you match a record in system A to the same thing in system B?
37. How far back does the history go? Was there a migration?
38. Who owns this system? Who do you call when it's down?
39. How often is it wrong, and how do you find out it was wrong?
40. Can you export this? Have you ever? What did you do with it?
```

CHANGE HISTORY AND APPETITE

TEXT

```
41. Has anyone tried to fix this before? What happened?
42. What changed in the last two years that made things better or worse?
43. What would have to be true for you to actually use something new?
44. What would make you refuse to use it?
45. Who on your team would be the hardest to convince, and why?
46. If I came back in six months and this had gone brilliantly, what would
    be different about your day?
```

CLOSING (NEVER SKIP THESE)

TEXT

```
47. What should I have asked you that I didn't?
48. What's the one thing you want to make sure gets written down?
49. Can you send me an example — a real one, redacted if needed?
50. Who else should I talk to? Can you introduce me?
51. Is it all right if I come back to you when I've got something to show?
52. I'll send my notes tomorrow — will you tell me what I got wrong?
```

✓ **Question 47 pays for the hour**

"What should I have asked you that I didn't?" routinely produces the single most valuable sentence of the interview. It costs nothing, signals humility, and lets the interviewee say the thing they have been waiting forty minutes to say. Ask it every time, then stay silent.

Questions to never ask

Do not ask	Why	Ask instead
"Would you use a tool that did X?"	Everyone says yes to hypothetical tools. Worthless signal.	"What do you do today when X comes up?"
"What features do you want?"	Makes them your product manager; you get a wish list, not a problem.	"Walk me through the last time this was painful."
"Don't you think it'd be better if…?"	Leading. They will agree to be polite and you will build it.	"How would that change what you do?"
"Why do you do it that way?"	Sounds like an accusation; triggers justification.	"What happens if you don't do that step?"
"Is the data clean?"	Always answered "yes" by people who have not looked.	"Show me three real records." Then look.
"How long does it take?"	Self-reported durations are wrong by 2-3x in both directions.	"Can we do one together now while I time it?"
"Do you have any problems?"	Too broad; invites "not really".	"What was the most annoying thing that happened this week?"

Silence, and other mechanics

COUNT TO FIVE

After they stop talking, say nothing for five seconds. It is excruciating and it works. Most people fill the gap with the qualification, the exception, or the honest version of what they just said.

ECHO THE LAST PHRASE

"...and then it goes to compliance." / "Compliance?" Repeating three words as a question is the lowest-effort way to go a level deeper without sounding like an interrogator.

ASK FOR THE ARTEFACT

"Can you show me?" beats any amount of description. Screens, spreadsheets, printed forms, sticky notes on monitors — the physical evidence of the real process.

TIMEBOX THE RANT

Let people vent for three minutes; it is both data and relationship-building. Then steer: "That's useful. Can you give me a specific example of when that bit you?"

INTERVIEW TWO, NOT SIX

Group interviews produce consensus, not truth. The quietest person in a six-person session knows the most and says the least. Prefer pairs or individuals.

STOP AT 55 MINUTES

Ending early builds more goodwill than any amount of extra content, and the follow-up meeting you get is better than the fifteen minutes you took.

Note-taking and verbatim quotes

Write what they said, not what you concluded. Your conclusion will change; their words will not. Use a two-column layout and mark quotes explicitly so that later you can tell evidence from inference.

TEXT

```
LIVE NOTE FORMAT (two columns, or two markers if typing linearly)

  " ... "  = verbatim, exact words, use sparingly and accurately
  >        = paraphrase of what they said
  ?        = my question (keeps the context of the answer)
  [[ ]]    = my inference / hypothesis - CLEARLY MARKED as mine
  TODO     = something to follow up
  NUM      = a number to verify against a system
  ART      = artefact they offered to send

Example:

? how does work arrive
> shared inbox, no assignment, "whoever gets there first"
"We don't have a system, we have a habit."                    ⟵ quote
NUM ~60 credit failures on Mondays, ~25 other days  [verify in SAP]
> half of them are ship-to vs credit-system address mismatch
"The old address thing. It's been like that since the merger."  ⟵ quote
[[ this looks deterministic - candidate for a matching rule, high volume ]]
ART Martin will send 10 real exception screenshots, redacted
TODO ask IT who owns the credit-check job
TODO ask Priya whether the merger data migration is documented
```

✦ **Collect quotes deliberately**

Three well-chosen verbatim quotes in a steering-committee deck do more persuasive work than ten charts. "We don't have a system, we have a habit" lands with an executive in a way that "process maturity is low" never will. Keep a running quote bank through the engagement, with attribution by role (never by name without permission), and deploy them at decision moments.

The interview write-up template

Write it the same day. Within 24 hours you lose roughly half of what was not written down, and almost all of the nuance.

TEXT

INTERVIEW NOTE

Who: Martin Hoeffner - Senior Order Management Specialist, 11 yrs
Date: 2026-02-04, 45 min, video, notes shared with him 2026-02-05
Purpose: Understand how order exceptions are detected and resolved
Present: Arun (interviewer), Sana (notes)

HEADLINE (one sentence)
About half of overnight credit-check failures are false positives caused by a ship-to/credit-system address mismatch, and clearing them manually before 11:00 is what stands between ~15 orders a day and a 24-hour delivery slip.

KEY FACTS (with verification status)
 F1 Overnight credit-check failures: ~25/day, ~60 on Mondays   [UNVERIFIED - self-reported; verify from SAP job log]
 F2 ~50% are address-mismatch false positives   [UNVERIFIED]
 F3 Manual override takes ~1 min each   [UNVERIFIED]
 F4 Orders not cleared by 11:00 miss the truck → +1 day   [VERIFIED - matches the DC cut-off in the logistics SOP]
 F5 No assignment rules on the shared inbox   [VERIFIED - observed on screen]

QUOTES
 "We don't have a system, we have a habit."
 "The old address thing. It's been like that since the merger."
 "If I'm off sick, nobody clears them. They just sit."

PAIN POINTS (ranked by his emphasis)
 1. Monday morning backlog - 60 items before any real work starts
 2. No visibility for anyone else; single point of failure (him)
 3. Having to open SAP, the credit tool and the shipping portal per case

OPPORTUNITIES SPOTTED   Est. size   Confidence
 O1 Auto-resolve address-mismatch false positives   ~3,900 orders/yr   Med
 O2 Assignment rules on shared inbox   unsized   High
 O3 Single-screen case view across 3 systems   ~11 min/case   Low

CONTRADICTS / CONFIRMS EARLIER INPUT
 - Contradicts Priya's statement that "credit checks are automated end to end"
 - Confirms Sunita's comment about Monday backlogs

RISKS / CONSTRAINTS LEARNED
 - Credit-check job owned by Finance IT, not Commercial IT. New stakeholder.
 - The merger data migration (2019) is apparently undocumented.

FOLLOW-UPS
 [ ] Pull SAP credit-block log, 60 days   Arun   by 07 Feb
 [ ] Martin to send 10 redacted exception examples   Martin by 06 Feb

```
  [ ] Identify owner of credit-check job              Sana    by 07 Feb
  [ ] Interview the Manila night shift lead           Arun    by 11 Feb

MY INFERENCES (not his words)
  [[ The real problem may be master-data divergence post-merger rather than
     the credit check itself. If so this is bigger and more valuable than
     the CS response-time brief we were given. Flag to Priya. ]]
```

Interview programme design

One interview is an anecdote. A programme is evidence. Plan the set, not just the sessions.

Layer	Who	How many	What you are getting
Sponsor	Budget holder, exec sponsor	1-2, repeated	Why now, success definition, political constraints
Management	Team leads, process owners	3-5	Official process, metrics, prior attempts
Practitioners	The people doing the work	6-10	Real process, workarounds, true volumes
Adjacent	Upstream and downstream teams	2-4	Handoffs, what they receive, what they resend
Technical	System owners, DBAs, integration team	2-4	Feasibility, access, data reality, release process
Dissenters	Known sceptics, previous project veterans	1-2	Failure modes, the thing nobody will say in the room

⚠ **Saturation, and the trap of only talking to managers**

Stop interviewing a layer when two consecutive sessions produce nothing new — that is saturation, usually at 5-8 people per role. But note that saturation among managers arrives fast and means nothing, because they share an official narrative. The practitioner layer is where variance lives, and it is the layer engineers skip because it is harder to schedule. If you have talked to four managers and no practitioners, you know the process as documented, which is to say you know fiction.

Observation and Process Mapping

People cannot accurately describe what they do. This is not dishonesty; it is how expertise works — once a task is automatic it stops being narratable, and the exceptions that consume most of the time feel too trivial to mention. Watching beats asking by a wide margin, and a process map built from observation rather than from a workshop whiteboard is the single most credible artefact you can put in front of a steering committee.

Shadowing: the protocol

1. **Get consent and explain the purpose clearly.**
   "I'm not assessing you, I'm assessing the system you have to work inside." Say it to them and, critically, to their manager, so nobody fears a time-and-motion study aimed at headcount.

2. **Sit beside, slightly behind, and shut up.**
   Watch at least one full cycle without interrupting. Write down questions rather than asking them.

3. **Ask in the gaps, not during the task.**
   When they finish an item: "you paused there — what were you checking?"

4. **Record every switch.**
   Application switches, tab switches, physical movements, phone calls, interruptions. The cost of a process is mostly in its seams.

5. **Observe across time, not just across people.**
   Monday morning, month-end, the night shift, peak season. A Tuesday-afternoon-only view of a seasonal process is a lie.

6. **Ask for the next three to be narrated aloud.**
   Think-aloud protocol surfaces decision rules — "I'd normally check the limit but this customer is fine" — that no documentation contains.

7. **Close by playing it back.**
   "So the steps are... did I miss anything?" They will correct you immediately and precisely.

TEXT

```
OBSERVATION LOG - raw capture format

Observer: Arun   Subject: Martin (Order Mgmt)   Date: 2026-02-06
Shift: Mon 08:00-10:00   Setting: desk, 2 monitors, headset
Consent: verbal, recorded in note. Manager informed 2026-02-05.

TIME   DURATION  ACTION                                    SYSTEM     NOTE
08:02   0:04:10  Opens shared inbox, scrolls, counts        Outlook    "sixty-two today
08:06   0:00:35  Opens first exception mail                 Outlook
08:07   0:01:10  Alt-tab to SAP, VA03, pastes order no     SAP        typed, not paste
08:08   0:00:48  Reads credit block reason                 SAP        squints, small f
08:09   0:01:30  Alt-tab to credit tool, searches customer CreditPro  *** re-login
08:10   0:00:20  Compares addresses on two screens         both       side by side, ma
08:11   0:00:25  Decides: false positive                   -          no rule written
08:11   0:00:55  Returns to SAP, releases block           SAP
08:12   0:00:15  Marks mail read, no reply                Outlook    no record kept
08:12   -        NEXT ITEM

INTERRUPTIONS: 08:19 phone 4 min; 08:34 colleague question 2 min;
               08:51 Teams ping from DC 6 min
RE-LOGINS:     3 in 2 hours (CreditPro session timeout = 15 min idle)
CYCLE TIMES:   n=19 items. min 2:10, median 4:05, max 14:30 (one needed a call)
BREAKDOWN of median 4:05
   finding/opening        0:40  (16%)
   SAP lookup             2:20  (57%)  ← includes 2 re-logins amortised
   decision               0:25  (10%)
   action                 0:40  (16%)

OBSERVED vs REPORTED: Martin said "about a minute each". Measured median
4:05. Not dishonesty - he times the decision, not the lookup.
```

The gap between reported and observed is the finding

"He said one minute, it is four" is not a gotcha — it is the most useful number you will produce that week, because every business case built on self-reported times is wrong in the same direction. Present it without blame: "self-reported estimates naturally exclude the lookup time, which is where the cost actually is."

Time and motion, honestly

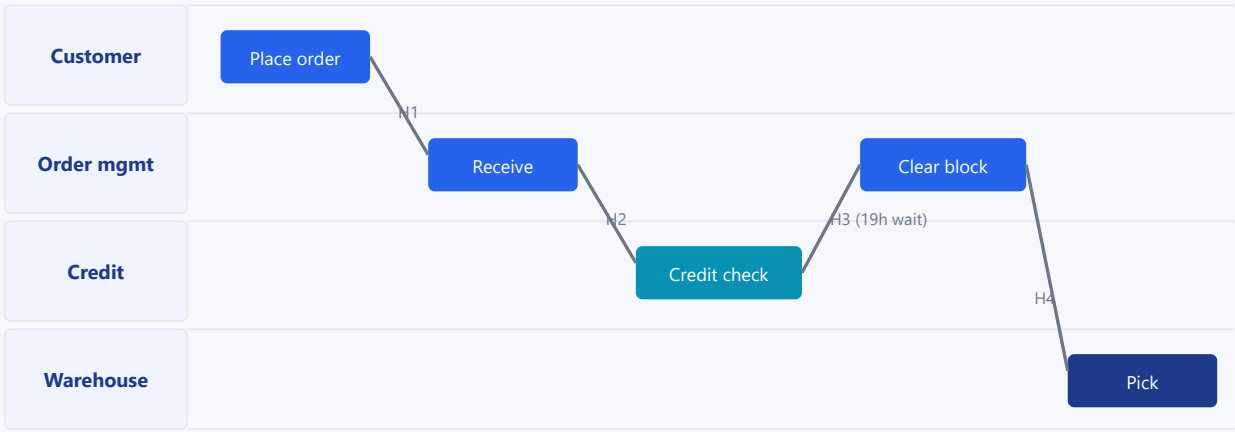
Technique	How	Good for	Watch out
Direct timing	Stopwatch n≥20 cycles, record the distribution not the average	Repetitive, bounded tasks	Observer effect; people speed up when watched
Work sampling	Random prompts across a week: "what are you doing right now?"	Mixed, interrupt-driven roles	Needs 100+ samples for stable proportions
System logs	Timestamps from the actual systems	Large n, no observer effect, free	Logs record system events, not human effort or waiting
Screen recording	Record a volunteer's session with consent	Precise step and switch counting	Consent and privacy; works-council rules in EU
Diary study	Participants log activity in a simple form for a week	Distributed teams you cannot visit	Compliance decays after day two; keep it to 5 fields

✕ **Never present timing data as individual performance**

The moment one person's numbers can be identified, every subsequent observation is theatre and your relationship with the floor is finished. Aggregate, anonymise, and say so loudly before you start. In Germany, France and much of the EU this is not just etiquette — systematic measurement of individual work rate may require works-council agreement. Ask HR before you time anything.

Swimlane maps

A swimlane shows who does what and, more importantly, where work crosses between actors. Every crossing is a queue, a potential loss, and a place where accountability is ambiguous. Count the crossings before you optimise any single step.



Four lane crossings. H3 — the return from the automated credit check to a human in order management — carries the 19-hour overnight queue. Draw the lanes before you draw the boxes.

TEXT

```
SWIMLANE IN TEXT (works in email, survives copy-paste, fast to build)

CUSTOMER   | (1) places order online
           |      |
ORDER MGMT |      v  (2) order lands in SAP, status=OPEN       [auto, 0 min]
           |      |
CREDIT SYS |      v  (3) nightly credit check runs 02:00       [auto, 0 min]
           |      |     -> PASS: status=RELEASED
           |      |     -> FAIL: status=BLOCKED, mail to shared inbox
           |      |
ORDER MGMT |      v  (4) ** WAIT UNTIL SOMEONE ARRIVES **      [0-14 h QUEUE]
           |      |  (5) human opens mail                      [0:40]
           |      |  (6) looks up order in SAP                 [1:20]
           |      |  (7) looks up customer in CreditPro        [1:00 + relogin]
           |      |  (8) compares ship-to addresses by eye     [0:25]
           |      |  (9) decides: false positive / real block  [0:25]
           |      |  (10) releases block in SAP                [0:40]
           |      |
WAREHOUSE  |      v  (11) picked in next wave (11:00 / 15:00)  [0-4 h QUEUE]

TOUCH TIME:  4 min 30 s
ELAPSED:     median 19 h 20 m
RATIO:       touch/elapsed = 0.4%
HANDOFFS:    4 lane crossings
DECISIONS:   1, undocumented, held in one person's head
```

Touch time versus elapsed time

Always report both, and always report the ratio. A process with 4 minutes of work spread over 19 hours is not a productivity problem, it is a queueing problem, and the fix is scheduling and routing, not making anyone faster. Executives who see "0.4%" immediately stop asking you to make the agents type quicker.

As-is to to-be, worked

TEXT

```
AS-IS  (measured, 2026-02)
  Steps performed by a human           7
  Systems touched                      3
  Lane crossings                       4
  Touch time                           4 m 30 s
  Elapsed, median                      19 h 20 m
  Elapsed, p90                         31 h
  Volume                               ~25/day, ~60 Mondays
  Decision rule                        undocumented, 1 person
  Failure if person absent             total; queue grows unboundedly
  Annual human cost                    ~7,000 items x 4.5 min = 525 h = 0.3 FTE
  Annual business cost                 ~3,900 delayed deliveries

TO-BE  (proposed, phase 1 - 6 weeks)
  (1) Credit check writes BLOCKED + reason code to a queue table, not mail
  (2) A rule evaluates: if block_reason = ADDRESS_MISMATCH
          AND normalised(ship_to) == normalised(credit_addr)
          THEN auto-release, write audit row
  (3) Anything else → assigned queue in the existing tool, round-robin,
      with a 2-hour SLA timer visible to the team lead
  (4) Single case view: order + credit + address diff on one screen

  Steps performed by a human           2 (only for true blocks)
  Systems touched                      1
  Lane crossings                       1
  Touch time (true blocks only)        1 m 40 s
  Elapsed, median                      projected 1 h 10 m
  Auto-resolved share                  projected 48% (from 300 labelled cases)
  Decision rule                        in code, reviewed, versioned
  Failure if person absent             degrades gracefully; queue visible
  Annual human saving                  ~390 h
  Annual business benefit              ~1,900 fewer delayed deliveries

WHAT WE DELIBERATELY DID NOT CHANGE
  - the nightly batch schedule (owned by Finance IT, 6-week change window)
  - the credit scoring logic itself (regulated, out of scope)
  - the warehouse pick waves

ASSUMPTIONS TO VALIDATE BEFORE COMMITTING
  A1 48% auto-resolve rate holds on a fresh sample  [test on 300 more cases]
  A2 Finance IT will expose the block reason code   [asked 07 Feb, no answer]
  A3 Audit accepts an automated release with a logged rule  [Legal, 2 weeks]
```

Finding waste without the jargon

You do not need to say "muda" to a customer. You do need to recognise eight patterns, because naming them turns a vague feeling of inefficiency into a list of removable things.

Pattern	What it looks like in an enterprise office	Typical fix
Waiting	Work sits in a queue, an inbox, a batch window, an approval	Routing, assignment, SLA timers, smaller batches
Transport	Data copied between systems, re-keyed, emailed as attachments	Integration, shared identifiers, a single source
Motion	Alt-tabbing between three systems to assemble one view	Composite screen, API aggregation
Over-processing	Checks nobody reads, reports nobody opens, four approvals	Delete the step; risk-tier the approvals
Defects	Rework caused by bad upstream data	Validate at entry, not downstream
Inventory	A backlog of 2,000 unprocessed cases that nobody triages	Age-based triage; stop the inflow first
Over-production	Generating all 400 reports because 6 people want 6 of them	Pull, not push; on-demand generation
Unused talent	Senior specialists spending 40% of time on clerical lookup	The most common and most saleable finding you will make

✓ **The highest-value question in process work**
"Why does this step exist?" asked about every step. A surprising proportion of enterprise steps exist because of a system limitation that was removed years ago, an audit finding from 2014 that was closed, or a person who left. Eliminating a step is worth more than automating it, costs nothing to implement, and requires only that someone with authority says yes.

Stakeholder Analysis

Projects do not fail in code review, they fail in a corridor conversation you were not part of. Stakeholder work is not politics-as-cynicism; it is the recognition that a solution only creates value if a specific set of named humans choose to adopt, fund and defend it. Mapping those humans — their power, their interest, their private incentives — is an engineering activity with the same rigour as capacity planning.

The six roles you must identify by name

ECONOMIC BUYER

Controls the money and can say yes alone. Often not in your meetings. If you cannot name them in week two, you have no project, you have an experiment someone is tolerating.

CHAMPION

Wants this to succeed for their own reasons and will spend political capital. Your job is to make them look good and arm them with material they can forward without editing.

BLOCKER

Has a reason — often legitimate — to stop you. Security, architecture review, a rival initiative, a budget line they own. Blockers are usually solvable; being surprised by one rarely is.

END USER

Must actually use the thing. Has almost no formal power and complete practical veto: quiet non-adoption kills more systems than any committee.

GATEKEEPER

Controls access to data, environments, people or the sponsor's calendar. Low formal rank, enormous ability to add six weeks. Treat with disproportionate respect.

TECHNICAL AUTHORITY

Must bless the design — enterprise architect, security officer, DPO. They cannot usually approve your project but they can absolutely stop it. Engage in week one, not at go-live.

The question that exposes a dead project

Ask your main contact: "If this works, whose budget pays for it next year?" If the answer is vague, hedged, or "we'd have to see", you are in a pilot that has no path to production and you should either get the economic buyer into a room within three weeks or manage your own expectations accordingly. A champion without a buyer is a pleasant six months leading nowhere.

Power/interest mapping



Position people by real power, not title. The CISO has high power and moderate interest — classic "keep satisfied": brief early, never surprise.

A worked stakeholder map

TEXT

STAKEHOLDER REGISTER – CS Response Time – v3 – 2026-02-18
(Internal working document. Never share. Never put it in a shared drive.)

NAME / ROLE	POWER	INT	STANCE	WHAT THEY WANT	WHAT THEY FEAR
Rachel Obi COO – ECONOMIC BUYER	HIGH	MED	Neutral	Cost line flat; the two QBR complaints to stop	Another IT programme that overruns
Access: via Priya only. Has 15 min at the monthly ops review. Move: Get one slide into Priya's March ops review with a measured before/after. Do not ask for a meeting yet – ask for a slide.					
Priya Menon Director, Cust Svc – CHAMPION	MED	HIGH	Advocate	Visible win before her Sept review; no new headcount ask	Being blamed for a failed pilot
Access: weekly 1:1, Tue 09:00. Move: Make her the presenter at steering. Draft her words. Never let her be surprised. Give her a number she can repeat from memory.					
Dr. Klaus Frey CISO	HIGH	LOW	Blocker	No new data flows outside the tenancy	A breach traced to a side project
Access: security review board, fortnightly, requires a submitted design. Move: Submit design to the 25 Feb board even though we are not ready. A conditional approval beats a late rejection. Offer to scope phase 1 entirely inside their tenancy to remove his objection.					
Sanjay Rao Head of Automation CoE – RIVAL	MED	MED	Hostile	Owns a competing RPA programme; wants this routed through his CoE	His programme being seen as slower/dearer
Access: peer, invited to working group. Move: Do NOT compete. Offer co-authorship: the auto-release rule runs as a component his CoE can reuse and claim. Give away the credit; keep the delivery. Escalating this would cost more than it wins.					
Martin Hoeffner Snr Order Mgmt – KEY USER + SME	LOW	HIGH	Advocate	Monday backlog gone; to not be the single point of failure	Being measured individually; headcount cut
Access: direct, daily. Move: Name him as process SME in the SOW. Show him every build before anyone else. His endorsement neutralises floor-level resistance.					
Anita Desai Finance IT – OWNS THE CREDIT JOB – GATEKEEPER	LOW	LOW	Neutral	To not be given more work	Owning support for something she didn't ask for
Access: ticket queue, 10-day SLA. THIS IS OUR CRITICAL PATH.					

```
  Move:   Escalate via Priya to her manager with a specific, tiny ask
          (expose one field). Make it a 2-hour job, not a project. Offer to
          write the change ourselves and have her review.

Works council rep    HIGH   LOW  Unknown  No individual         Surveillance;
(DE site)                                 performance tracking  job loss
  Access: formal, via HR, 3-week lead time.
  Move:   Engage NOW, before any measurement. Frame as removing a Monday
          backlog, not as productivity measurement. Commit in writing to
          aggregate-only metrics. Getting this wrong stops the project dead.

TOP THREE RISKS FROM THIS MAP
 R1 Anita's 10-day ticket SLA is on our critical path. Mitigate by week 3.
 R2 CISO approval is fortnightly; missing 25 Feb costs 2 weeks.
 R3 Sanjay can reframe this as duplicating his CoE at any steering meeting.
    Pre-wire him before every steering committee, without exception.
```

⚠ **Keep this document off the customer's systems**

A stakeholder register with the words "hostile" and "blocker" next to named employees is radioactive if it leaks. Keep it in your own notes, in your own storage, never in the customer's SharePoint, never attached to an email, never on a shared screen. If you must discuss it, do so verbally. Many consultants have ended engagements by screen-sharing the wrong tab.

Influence strategy by role

Role	What they respond to	What to send them	Cadence
Economic buyer	Money, risk, time, peer comparison	One slide: before, after, cost, date	Monthly, or at decision points only
Champion	Being made to look competent	Forwardable updates written in their voice	Weekly 1:1, no exceptions
Blocker	Their own criteria being met explicitly	A design doc addressing their checklist line by line	Early, then at each gate
End user	Their day getting better; being consulted	Working software, early and ugly	Continuous; demo every week
Gatekeeper	Small, specific, low-effort asks	A precise request with the work pre-done	As needed, with long lead time
Technical authority	Rigour, precedent, standards compliance	Architecture doc with their standards referenced	Week one, then at design freeze
Rival	Credit, inclusion, non-threat	Co-authorship, shared components, their logo on it	Before every public forum

The engagement plan

TEXT

```
ENGAGEMENT PLAN - February

WHO              WHAT                                 WHEN         OWNER   DONE
Priya            Weekly 1:1, 30 min                   Tue 09:00    Arun    recurring
Priya            Pre-wire steering deck               Mon before   Arun
Rachel (COO)     1 slide in her ops review            03 Mar       Priya
Klaus (CISO)     Design submission                    by 20 Feb    Arun
Klaus (CISO)     Security review board                25 Feb       Arun
Sanjay           Coffee, no agenda, before SteerCo    12 Feb       Arun
Sanjay           Invite to working group              weekly       Arun
Martin + 2       Demo of whatever exists              Thu 15:00    Arun    weekly
Anita (Fin IT)   Escalated ask for block-reason       by 14 Feb    Priya
Works council    Intro via HR, framing conversation   by 19 Feb    Priya
All users (22)   Written FAQ: "what this means        by 28 Feb    Arun
                 for your job"

RULE: nobody hears about this project first from a steering committee deck.
Everyone on this list hears it from a human, before it is written down.
```

Reading the room

SIGNALS YOU ARE WINNING

They start inviting you to meetings you did not ask for. People forward you things unprompted. Someone uses your terminology in a meeting you are not running. A stakeholder defends your project when you are not present. You get asked about something out of scope — that is trust.

SIGNALS YOU ARE LOSING

Meetings get shorter or moved. Your champion stops replying within a day. Decisions are "still being discussed". Someone new appears in the governance chain. You are asked for a "business case refresh" in month four. You hear a project update from a third party.

✦ **When the signals turn bad, act immediately and directly**

Do not wait and hope. Ask your champion, face to face: "I've noticed the steering meeting has moved twice and I'm not sure we still have momentum. What's changed that I'm not seeing?" People almost always tell you. The alternative — discovering in month five that the budget moved in month three — is unrecoverable.

RACI, Governance and Operating Cadence

Governance is unfashionable among engineers and indispensable on enterprise engagements, because it is the machinery that converts "someone should decide this" into a decision with a date and a name. Light governance done consistently beats heavy governance done performatively. The goal is a small number of forums, each with one job, each with a written output.

RACI without the ceremony

Responsible does the work. Accountable is the single name who answers for the outcome and can approve. Consulted is asked before. Informed is told after. One A per row, always; two A's means nobody is accountable.

TEXT

```
RACI — CS Response Time, Phase 1

                                  Arun  Priya Martin Anita Klaus Sanjay
Discovery and process mapping      R     A     C      I     I     I
Problem statement sign-off         R     A     C      -     -     C
Solution design                    R     C     C      C     A     C
Security design approval           R     I     -      -     A     -
Credit-system field exposure       C     C     -      R/A   I     -
Build the auto-release rule        R/A   I     C      I     I     C
Test data provision                C     A     R      C     -     -
UAT sign-off                       C     A     R      -     -     -
Go-live decision                   C     A     C      C     C     I
Hypercare (2 weeks)                R/A   I     C      I     -     -
Operational handover               R     C     A      C     I     C
Benefit measurement                R     A     C      -     -     I

RULES THAT MAKE THIS USEFUL
 - If a row has no A, that work will not happen. Find the A or cut the row.
 - If a row has two A's, there will be a fight. Resolve it on paper now.
 - If one person is A on everything, they are a bottleneck. Spread it.
 - "Operational handover: Martin is A" is the line that makes the handover
   real. Without it, handover is a document nobody owns.
 - Review the RACI at phase boundaries. It goes stale within a month.
```

The three-forum model

Forum	Who	Frequency	Only job	Output
Steering committee	Sponsor, buyer, key function heads, you	Monthly, 45 min	Decisions, escalations, funding, scope changes	Decision log entry with owner and date
Working group	Process owner, SMEs, technical owners, you	Weekly, 60 min	Unblock, agree detail, review increments	Action list, open questions with owners
Standup	Delivery team only	Daily, 15 min	Coordination and blockers	Nothing written beyond the board

> ⚠ **The most common governance failure**
> Using the steering committee as a status meeting. If you spend forty minutes presenting progress and five minutes on decisions, you have wasted the most expensive hour in the project. Send status in writing 48 hours before. Open the meeting with "I need three decisions today" and list them on the first slide. Senior people will thank you and will attend the next one.

Steering committee pack and agenda

TEXT

```
STEERING COMMITTEE – 45 MINUTES – AGENDA

Sent 48h in advance with the pack. Pack is 5 slides maximum.

00:00  Status in one slide                                  [3 min, no discussion]
       RAG, what shipped since last time, what is next.
       "Questions on status – hold them, I'll take them at the end."

00:03  DECISION 1: Scope of phase 2                         [10 min]
       Recommendation: include returns automation, defer phone channel.
       Options, cost, and what I recommend. I need a yes/no today.

00:13  DECISION 2: Finance IT resource for the field change [10 min]
       Ask: 2 days of Anita's time before 28 Feb.
       If no: phase 1 slips 3 weeks. That is the trade.

00:23  DECISION 3: Works council engagement approach        [10 min]
       Recommendation: HR-led framing session before any measurement.

00:33  Risks requiring your attention (top 3 only)          [7 min]

00:40  AOB and confirm decisions                            [5 min]
       Read back each decision, owner, date. Everyone hears it.

00:45  End on time, always.

AFTER: decision log entry circulated within 2 hours. Not 2 days.
```

TEXT

```
DECISION LOG (one file, append-only, shared with the customer)

ID    DATE        DECISION                          MADE BY   STATUS  NOTE
D-001 2026-02-04  Scope phase 1 to queue time only   Priya     Agreed  See problem
                  (not draft quality)                                 statement v2
D-002 2026-02-11  Build auto-release rule in our     Klaus     Agreed  Conditional:
                  tenancy, read-only to SAP                           pen test first
D-003 2026-02-18  Defer returns automation to        Rachel    Agreed  Revisit Apr
                  phase 2
D-004 2026-02-18  Finance IT to expose block_reason  Rachel    Open    Owner: Priya
                  field - 2 days of Anita's time                      Due 28 Feb
D-005 2026-02-25  Aggregate-only metrics; no         Priya     Agreed  Committed in
                  individual productivity reporting                   writing to WC

WHY APPEND-ONLY: in month six somebody will say "we never agreed to that".
This file, with dates and names, ends that conversation in ten seconds.
Never edit a past row; add a superseding row and reference the old ID.
```

Working group agenda

TEXT

```
WORKING GROUP - 60 MIN - WEEKLY

00:00  Demo whatever exists, even if broken          [10 min]
       Rule: something runs on screen every single week. No slides.
00:10  Decisions needed from this group              [15 min]
       Detail-level: field definitions, edge-case rules, naming.
00:25  Open questions - walk the list                [15 min]
       Each has an owner and a date. Chase the overdue ones here.
00:40  Blockers and escalations                      [10 min]
       Anything that needs the steering committee gets drafted now.
00:50  Next week: what I will show you               [5 min]
00:55  Buffer / overflow                             [5 min]

STANDING ARTEFACT: the open-questions list. It is the heartbeat of the
engagement. If it has fewer than 5 items you are not asking enough; if it
has more than 25 you are not closing them.
```

TEXT

```
OPEN QUESTIONS LIST

ID   QUESTION                                      OWNER   ASKED   DUE    STATUS
Q-14 What is the exact block_reason code for       Anita   06 Feb  14 Feb OVERDUE
     an address mismatch? Is it stable?                                    (chase)
Q-19 Do we need to notify the customer when a      Legal   11 Feb  25 Feb Open
     block is auto-released?
Q-22 What is the address normalisation standard    Martin  12 Feb  19 Feb Open
     - do we strip "Str." vs "Strasse"?
Q-23 Who approves releasing a block over EUR       Priya   12 Feb  19 Feb Answered
     50,000 - is there a value threshold?                                  (no
                                                                           threshold)
Q-25 Does the works council agreement cover        HR      18 Feb  04 Mar Open
     queue-level SLA timers?

RULE: every question has an owner who is a named human, and a date. "The
team" is not an owner. "Soon" is not a date. Overdue items get read aloud
in the working group, which is remarkably effective at closing them.
```

✓ **Cadence is a trust machine**

The specific rituals matter far less than their absolute reliability. A mediocre status email that arrives every Friday at 16:00 for nine months builds more confidence than a brilliant one that arrives when you feel you have something to say. Pick a cadence you can sustain on your worst week, and then never miss it — including, especially, the weeks when the news is bad.

Scoping and the Statement of Work

The statement of work is the document that decides whether month five is a collaboration or an argument. Engineers under-invest in it because it feels like paperwork; experienced FDEs treat it as the primary design artefact, because every ambiguity you leave in it will be resolved, later, in the direction that costs you. Write it as though a stranger will read it in six months to adjudicate a dispute — because that is exactly what happens.

Deliverables versus outcomes

	Deliverable	Outcome
Example	"An auto-release rule deployed to production"	"Median response time below 6 hours"
You control it	Yes	Partially
Customer values it	Only as a means	Yes, this is what they buy
Safe to contract on	Yes	Only with named dependencies
Risk	You deliver artefacts nobody uses, and are technically compliant	You are held to a result that depends on their adoption

The resolution is to contract on deliverables and *declare* the outcome with its dependencies. Write the outcome in the SOW under a heading that explicitly states what must be true on the customer's side for it to be achieved. You get alignment on purpose without accepting liability for other people's behaviour.

TEXT

```
HOW TO WRITE IT

INTENDED OUTCOME
  Median first-response time for customer emails falls from 26 hours to
  below 6 hours, measured on the mail-server/CRM join defined in Appendix B,
  sustained over 4 consecutive weeks.

DEPENDENCIES ON WHICH THAT OUTCOME RELIES (customer-owned)
  - The assigned-queue workflow is adopted by all 9 CS agents
  - The Finance IT block_reason field is exposed by 28 Feb
  - Volume does not increase by more than 20% over the baseline period
  - No change freeze is imposed during the measurement window

DELIVERABLES (supplier-owned, these are what acceptance tests)
  D1 Auto-release rule, deployed, with audit logging      [acceptance: A1]
  D2 Assignment rules configured in the existing tool     [acceptance: A2]
  D3 Composite case view (order + credit + address diff)  [acceptance: A3]
  D4 Runbook and handover pack                            [acceptance: A4]
  D5 Two training sessions, recorded                      [acceptance: A5]
```

The full annotated SOW skeleton

TEXT

```
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STATEMENT OF WORK
Customer:   Nordstahl Distribution GmbH
Supplier:   [Your company]
Project:    Customer Service Response Time – Phase 1
SOW ref:    NSD-2026-014   Version: 1.3   Date: 2026-02-20
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1. BACKGROUND                                  [why this exists, 1 para]
   Nordstahl's customer-service team responds to inbound customer email
   in a median of 26 hours against a 4-hour commitment in its customer
   charter. Analysis conducted 03–14 Feb 2026 (ref: Discovery Findings
   v2) attributes 19 of those 26 hours to queue time rather than to
   agent handling.
   >> ANNOTATION: anchor to a dated, shared document. This paragraph is
      what a dispute resolver reads first. Keep it factual, no selling.

2. OBJECTIVES                                  [outcomes, with numbers]
   2.1 Reduce median first-response time to below 6 hours.
   2.2 Eliminate the Monday morning exception backlog.
   2.3 Remove the single-person dependency for exception clearing.
   >> ANNOTATION: three is the right number. Ten objectives means none.

3. SCOPE OF WORK                               [what we will do]
   3.1 Analysis of credit-block exception volumes and causes
   3.2 Design and build of an auto-release rule for address-mismatch
       false positives, with full audit logging
   3.3 Configuration of assignment rules in the existing CS tool
   3.4 A composite case view combining SAP order data, CreditPro
       customer data and an address comparison
   3.5 Documentation: solution design, runbook, handover pack
   3.6 Training: two 90-minute sessions, recorded, plus a quick-reference
   3.7 Two weeks of hypercare following go-live
   >> ANNOTATION: each item must be independently verifiable. "Improve
      the process" is not a scope item. "Provide support" is not either –
      say how much, for how long, at what response time.

4. OUT OF SCOPE                                [the most valuable section]
   The following are explicitly NOT included and would require a change
   request and additional fees:
   4.1 Any modification to SAP configuration, code or master data
   4.2 Changes to the credit scoring logic itself
   4.3 The telephone channel
   4.4 Draft-quality, tone or content of agent replies
   4.5 The credit-note approval workflow
   4.6 Migration or cleansing of historical customer master data
   4.7 Integration with any system not named in section 3
   4.8 Support outside 09:00–18:00 CET, Mon–Fri
```

4.9 Any work at the Pune or Manila sites
4.10 Production support beyond the 2-week hypercare period
>> ANNOTATION: write this section LONGER than you think necessary. Every line here is an argument you will not have in May. When a customer asks "why is this list so long?", answer honestly: "so that neither of us is surprised. If any of these should be in scope, now is the cheapest time to add them."

5. DELIVERABLES AND ACCEPTANCE CRITERIA

ID	Deliverable	Acceptance criterion	Due
D1	Auto-release rule	A1: Processes 300 historical cases; ≥95% agreement with a manually labelled gold set; 0 false releases on the 40 true-block cases; audit row written for every decision	14 Mar
D2	Assignment rules	A2: 20 test messages route to the correct agent queue; SLA timer visible to team lead	07 Mar
D3	Composite case view	A3: Displays order, credit status and address diff for 10 nominated test orders in <3 s	21 Mar
D4	Runbook + handover	A4: Named Nordstahl operator completes 3 scenarios from the runbook unaided, observed	28 Mar
D5	Training	A5: 2 sessions delivered, ≥7 of 9 agents attended, recording provided	28 Mar

>> ANNOTATION: acceptance criteria must be testable by someone who was not in the room. "To the customer's satisfaction" is not an acceptance criterion, it is an open invoice. Note A4: the test is that THEIR person can do it, not that you wrote a document.

6. ACCEPTANCE PROCESS
6.1 Supplier notifies completion in writing.
6.2 Customer has 5 working days to test against the stated criteria.
6.3 Rejection must be in writing citing the specific criterion failed.
6.4 Absent written response within 5 working days, the deliverable is deemed accepted.
>> ANNOTATION: 6.4 is the clause that prevents indefinite limbo. Fight for it. Without deemed acceptance, "we haven't got round to testing" can hold an invoice for a quarter.

7. ASSUMPTIONS [if these are false, cost changes]
7.1 Customer provides read access to the SAP credit-block tables by 28 Feb 2026.
7.2 Finance IT exposes the block_reason field by 28 Feb 2026.
7.3 Security review board approval is obtained at the 25 Feb sitting.
7.4 Named SME (M. Hoeffner) is available 4 hours per week.

```
   7.5 Volumes remain within +/-20% of the Jan 2026 baseline.
   7.6 No change freeze between 01 Mar and 31 Mar 2026.
   7.7 Works council consultation, where required, is led by the Customer
       and completed by 28 Feb 2026.
   7.8 Existing CS tool supports assignment rules without a licence upgrade.
   >> ANNOTATION: each assumption must name a date and an owner. Then add
      the consequence clause below - this is what turns assumptions from
      decoration into a working mechanism.

8. CONSEQUENCES OF ASSUMPTION FAILURE
   If any assumption in section 7 is not met by its stated date, the
   Supplier will notify the Customer in writing within 2 working days
   with the revised impact on timeline and cost. Delay caused by
   unmet assumptions extends the schedule day-for-day.
   >> ANNOTATION: without this clause, assumptions are wishes. With it,
      your week-6 slip conversation is a contractual formality rather
      than an apology.

9. CUSTOMER RESPONSIBILITIES
   9.1 Nominate a single point of contact with decision authority (Priya
       Menon) available weekly.
   9.2 Provide workspace/VPN access for Supplier personnel by 25 Feb.
   9.3 Provide test data per Appendix B within 5 working days of request.
   9.4 Make agents available for 2 training sessions.
   9.5 Own all internal communications to staff and works council.
   >> ANNOTATION: name a single decision-maker. Projects with "the
      steering committee decides" at this line move at one decision
      per month.

10. COMMERCIAL MODEL
    Time and materials, capped at EUR 96,000.
    Rate: EUR 1,200 per consultant-day.
    Estimated effort: 80 consultant-days (see Appendix C).
    Invoiced monthly in arrears against a timesheet.
    The Supplier will notify the Customer when 70% of the cap is consumed.
    >> ANNOTATION: the 70% notification clause is non-negotiable and
       protects both sides. The cap makes T&M acceptable to procurement.

11. CHANGE CONTROL
    Any change to sections 3, 4, 5, 7 or 10 requires a written change
    request signed by both parties, stating the change, the impact on
    cost, the impact on schedule and the impact on other deliverables.
    No work on a change begins before signature.
    >> ANNOTATION: "no work begins before signature" is the sentence that
       prevents scope creep. You will be asked to breach it in week 3.
       Decline, politely, every time, and the SOW stays meaningful.

12. TEAM AND KEY PERSONNEL
    Supplier: A. Krishnan (lead engineer, 1.0 FTE), S. Ali (0.4 FTE)
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```
    Customer: P. Menon (sponsor), M. Hoeffner (SME, 4h/wk)

13. TIMELINE AND MILESTONES
    M1 Design approved and security sign-off        28 Feb
    M2 Auto-release rule in test                    14 Mar
    M3 Composite view in test                       21 Mar
    M4 Go-live                                      28 Mar
    M5 Hypercare ends, handover accepted            11 Apr

14. DEPENDENCIES ON THIRD PARTIES
    14.1 Finance IT (internal to Customer) - block_reason field
    14.2 CreditPro vendor - API rate limit increase to 60 rpm
    >> ANNOTATION: third-party dependencies are the most common cause of
       slippage and the one customers forget they own. Name them.

15. RISKS (top 3, with owner and mitigation)
    R1 Finance IT capacity (Customer-owned)
    R2 Security approval timing (Shared)
    R3 Auto-release accuracy below 95% on fresh data (Supplier-owned;
       mitigation: fall back to assisted review, scope unchanged)

16. INTELLECTUAL PROPERTY, DATA AND CONFIDENTIALITY
    [per master services agreement, ref MSA-2024-07, clauses 8-11]
    Customer data remains Customer's. Supplier retains rights to generic
    methods, tools and know-how not embodying Customer data.

17. SIGNATURES
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The clause engineers forget

Section 8 — consequences of assumption failure. Without it, you will list eight assumptions, three will fail, you will absorb the delay silently to stay likeable, and in month four you will be three weeks late with no documented reason. With it, each failure triggers a two-day written notice and the schedule moves formally. This single clause has saved more engagements than any technical practice.

Time and materials versus fixed price

	Time and materials	Fixed price	T&M with cap
Risk sits with	Customer	Supplier	Supplier above the cap
Suits	Discovery, unclear scope, R&D	Well-understood, repeatable work	Most FDE engagements
Scope discipline	Weak — change is easy, which is good and bad	Strong — every change is a negotiation	Moderate; the cap forces conversations
Procurement comfort	Low; looks open-ended	High	High
Failure mode	Endless engagement, value drift, budget fatigue	Supplier cuts corners, dispute over "included"	Cap consumed at 60% of the work
Required discipline	Weekly burn reporting	Ruthless out-of-scope list	70% notification, honest burn

⚠ **Never accept fixed price on an undefined problem**
If you are being asked for a fixed price before discovery is done, the correct answer is a two-phase structure: a small fixed-price discovery (two to four weeks, defined output: a findings document, a design and a fixed-price proposal for phase 2), followed by a priced build. This is a standard, respectable commercial pattern, it protects both parties, and customers who refuse it are telling you they want you to absorb their uncertainty.

Estimation techniques that survive contact

1. **Decompose to units of 0.5–3 days.**
Anything estimated at "two weeks" is an un-decomposed lump concealing unknowns. If you cannot break it down, that is a research task, not a build task — estimate the research.

2. **Three-point estimate per item.**
Optimistic, likely, pessimistic. Expected = (O + 4L + P) / 6. The spread is more informative than the midpoint: items where P is 4× O are your real risks.

3. **Apply a discovery multiplier honestly.**
On a first engagement with a new customer, multiply by 1.5–2.0 for access delays, environment surprises, and the three meetings per week you did not plan. This is not padding; it is the

measured cost of working inside someone else's organisation.

4 **Estimate the non-code work explicitly.**
Meetings, documentation, demos, security review, test data chasing, training. On enterprise engagements this is 30–50% of total effort and engineers routinely cost it at zero.

5 **Record your estimates and review them.**
After each milestone compare estimate to actual per item. Within three engagements you will know your personal multiplier, which is worth more than any estimation framework.

6 **Never estimate in a meeting.**
"I'll come back to you tomorrow with a number" is always the right answer. Numbers given verbally under social pressure become commitments.

TEXT

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ESTIMATE WORKSHEET — Appendix C

ITEM                                    O    L    P    E=(O+4L+P)/6
Data access + environment setup         2    4   10     4.7
Pull + label 300 historical cases       2    3    5     3.2
Address normalisation rule design       1    3    8     3.5
Auto-release rule build                 3    5    8     5.2
Audit logging + reversal path           1    2    4     2.2
Assignment rules config                 1    2    5     2.3
Composite case view (3 sources)         4    7   14     7.7
Integration testing                     2    4    7     4.2
Security review prep + remediation      2    4   10     4.7
Runbook + handover pack                 2    3    5     3.2
Training prep + 2 sessions              1    2    3     2.0
Hypercare (2 weeks, 40% loading)        4    4    4     4.0
                                                      ------
SUBTOTAL build                                         46.9

NON-BUILD LOADING
  Working group + steering (1.5 h/wk x 8)               1.5
  Weekly demo prep                                      2.0
  Status reporting and decision logs                    1.5
  Stakeholder meetings not otherwise counted            4.0
  Travel (2 site visits)                                2.0
                                                      ------
SUBTOTAL non-build                                     11.0

Raw total                                              57.9
New-customer multiplier x1.35                          78.2
ROUNDED ESTIMATE                                       80 days

HIGHEST-RISK ITEMS (P/O ratio > 3): data access (5.0),
security review (5.0), address normalisation (8.0).
>> Those three are where the cap gets consumed. Front-load them.
```

Running a Discovery Workshop

A workshop is an expensive instrument — twelve people for three hours is roughly a consultant-week of salary — and it is the wrong tool for gathering information you could get in interviews. Use a workshop for the three things only a group can do: build shared understanding, make collective trade-offs visible, and produce a prioritised list that everyone in the room has publicly endorsed. Anything else is a meeting with a flipchart.

Design principles

DECIDE THE OUTPUT FIRST

Write the artefact you intend to leave with — "a ranked list of 8 opportunities with owners" — before designing a single exercise. Every activity either produces it or gets cut.

INTERVIEW BEFORE YOU WORKSHOP

Do your one-to-ones first. Walking in with a pre-built straw man draft of the findings makes the room far more productive than starting from a blank wall.

DIVERGENT THEN CONVERGENT

Never mix. Generate broadly with no criticism, then filter against explicit criteria. Rooms that evaluate while generating produce four safe ideas.

EVERYONE WRITES

Silent individual generation before any discussion. It cancels seniority bias and triples idea count. This single technique is the biggest quality lever in facilitation.

TIMEBOX VISIBLY

Put a timer on the screen. It depersonalises the interruption — the clock is cutting you off, not the facilitator.

CAPTURE AS YOU GO

Nothing goes on a sticky note that is not photographed and transcribed the same day. Workshop output that is not written up within 24 hours never gets written up.

A full worked three-hour agenda

TEXT

```
DISCOVERY WORKSHOP - CUSTOMER SERVICE RESPONSE TIME
3 hours | 11 participants | Thursday 13 Feb, 09:00-12:00, Room 4.02 + Teams

OBJECTIVE (stated on the wall all morning)
"Leave with a ranked list of opportunities, a named owner for each of the
top five, and agreement on what phase 1 contains."

PARTICIPANTS
  Priya Menon        Director CS            sponsor, must be present
  Martin Hoeffner    Order Mgmt SME         key knowledge
  Sunita Raghavan    CS Team Lead
  3 x CS agents                             the people who do the work
  Anita Desai        Finance IT             gatekeeper - 1 h only, 10:00-11:00
  Sanjay Rao         Automation CoE         rival - MUST be invited
  Thomas Berg        Enterprise Architect
  Arun               facilitator
  Sana               scribe (does not facilitate)

PRE-READ (sent Monday, 2 pages, with a one-line ask)
  - The measured baseline: 26h median, breakdown by stage
  - 6 anonymised quotes from interviews
  - The 11 draft opportunities, unranked
  - ASK: "come having read this, and bring one thing you think we've
    missed. 10 minutes of prep, no deck needed."

MATERIALS
  Printed process map (A0, on the wall), sticky notes in 3 colours,
  dot stickers (5 per person, 2 colours), markers, timer on screen,
  Miro board mirrored for the 2 remote attendees

_____________________________________________________________________

09:00  WELCOME AND FRAME                                   [10 min]
       - Why we're here, what we will leave with
       - "Three rules: everyone writes before anyone talks; no solutions
         before 10:15; if it's not on a sticky it didn't happen."
       - Round the room: name, role, one word for how CS feels today
         (gets every voice heard in the first 10 minutes - critical)

09:10  THE BASELINE - what the data says                   [15 min]
       - 4 slides max: volume, median, the 4-stage breakdown, the Pareto
       - Three verbatim quotes read aloud
       - "Does anything here not match your experience?" → capture
         disagreements on stickies, do NOT debate them now
       >> FACILITATOR NOTE: this is where resistance surfaces. Welcome it
          warmly, write it down, move on. "Great - that's exactly the kind
          of thing I need. On a sticky, we'll come back to it."

09:25  WALK THE PROCESS MAP                                [25 min]
```

```
       - A0 map on the wall, walk it step by step
       - At each step: "what actually happens here that isn't on this map?"
       - Participants add stickies directly onto the map
       - RED stickies = pain, YELLOW = workaround, BLUE = question
       >> This exercise does the most work in the whole session. The map
          being visibly WRONG is the point - people cannot resist correcting
          a wrong diagram, and correction is the richest form of input.

09:50  BREAK                                                  [10 min]
       >> Never skip. The corridor conversations in this 10 minutes are
          where you learn who really objects to what. Stand near the coffee.

10:00  SILENT GENERATION - "what would make this better?"     [10 min]
       - Everyone writes, alone, one idea per sticky, as many as they like
       - No talking at all. Timer visible.
       - Anita joins here (her 1 hour starts now, deliberately placed so
         she is present for the feasibility discussion)

10:10  AFFINITY CLUSTERING                                    [20 min]
       - Everyone puts stickies on the wall, reads others', groups them
       - Facilitator names clusters out loud and checks: "is this cluster
         'routing and assignment'? Does everyone agree with that name?"
       - Expect 7-12 clusters from ~60 stickies
       >> Let participants do the physical grouping. Facilitator-led
          clustering imposes your categories and halves the buy-in.

10:30  SIZING EACH CLUSTER - fast, rough, public              [25 min]
       - For each cluster, 2 minutes: "roughly how many cases a week?
         roughly how long each? who owns the system?"
       - Write the number on the cluster. Mark guesses with "?"
       - Anita confirms or kills feasibility on the three that touch
         Finance IT systems - this is why she is in the room
       >> Rough numbers out loud in a group are self-correcting. Someone
          always says "no, it's nothing like that many."

10:55  DOT VOTING                                             [10 min]
       - 3 GREEN dots each = "biggest impact on our customers"
       - 2 BLUE dots each = "easiest for us to actually do"
       - Vote silently and simultaneously, all at once, no discussion
       - Sponsor votes LAST and is told her dots count the same
       >> Simultaneous voting prevents anchoring. If the director votes
          first, everyone else's dots cluster around hers.

11:05  THE IMPACT/EFFORT PLOT                                 [20 min]
       - Plot each cluster using the two dot counts
       - Discuss ONLY the disagreements: items with high green and high
         blue variance. "Four people think this is easy and three think
         it's very hard. Who thinks it's hard, and why?"
       >> The variance is the signal. Consensus items need no discussion;
```

```
          disputed items are where the hidden constraint lives.

11:25  THE "ONE THING BY FRIDAY" TEST                        [10 min]
       - "If we could only do one thing, and it had to be live by a week
         on Friday, which one?"
       - Forces a real priority rather than a polite ranking
       - Expect the answer to differ from the dot vote. That difference
         is worth discussing for five minutes.

11:35  OWNERS AND NEXT STEPS                                 [15 min]
       - Top 5 clusters: name an owner for each, out loud, in the room
       - "Martin, are you happy to own the address-matching rule
         definition? What do you need from me by when?"
       - Capture on screen so everyone sees their name being typed
       >> Owners assigned in a room, verbally, in front of peers, stick.
          Owners assigned by email afterwards do not.

11:50  CLOSE                                                 [10 min]
       - Read back: top 5, owners, dates
       - "What did we not talk about that we should have?"
       - What happens next and when they'll hear from me (be specific:
         "write-up in your inbox by 17:00 tomorrow")
12:00  END ON TIME
_____________________________________________________________________

WITHIN 24 HOURS: photographs of every wall, full transcription, the
ranked list, owners, and the open questions, in one email to all 11.
```

Facilitation techniques

Situation	Move	Exact words
One person dominating	Structural, not personal — change the format	"Let's hear from people who haven't spoken yet. Sunita, what's your read?"
Still dominating	Give them a job that requires listening	"Sanjay, would you capture the themes on the board for this section?"
Still dominating	Private, in the break, direct and warm	"You've got the most context in the room and I need the quieter people to talk. Can you hold back for twenty minutes and I'll come to you?"
Silence after a question	Wait. Then narrow the question	[five seconds] "Let me make it easier — Martin, what happened on Monday?"
Two people arguing	Find the shared fact, park the rest	"You agree on what happens, you disagree on how often. Let's measure it — who can pull that number by Tuesday?"
Rat-holing on detail	Park it visibly	"That's important and too detailed for this group. Parking it — Sana, on the list with Thomas as owner."
Sponsor derails the agenda	Accept and re-contract the time	"Happy to spend twenty minutes on that. It means we drop the sizing exercise. Is that the trade you want?"
Room goes negative	Acknowledge fully, then redirect to control	"You've tried this twice and it failed. What would have to be different this time?"
Nobody will criticise	Anonymise the channel	Silent stickies, or "write down the reason this will fail — don't sign it"
Remote people silent	Call on them by name, first, every round	"Before we go round the room — Klaus, Rahul, what are you seeing?"

✦ **The pre-mortem**

Fifteen minutes that outperforms any risk workshop. "It is September. This project has failed badly. Write down, individually and silently, why." People will state risks in that frame that they would never raise as an objection, because criticising a hypothetical past failure is socially safe in a way that criticising a live plan is not. You will get the political risks, not just the technical ones.

Remote versus in-person

Aspect	In-person	Remote adjustment
Duration	3 hours with one break	Two 90-minute sessions on separate days; never 3 hours on video
Generation	Stickies on a wall	Miro/FigJam with private mode on, or a shared doc with named sections
Voting	Dot stickers	Simultaneous reveal — everyone types a number in chat, submit on three
Side conversations	Happen in the break; eavesdrop	Lost entirely. Compensate with 1:1 calls afterwards.
Reading the room	Body language	Cameras on as a stated norm; watch for people who stop nodding
Dominance	Visible, manageable	Worse. Use strict round-robin and named invitations.
Hybrid	—	The worst format. If you must: a dedicated remote-advocate facilitator, and all artefacts digital so the room cannot work on a wall the remote people cannot see.

⚠ **Hybrid workshops fail by default**

In a hybrid session the people in the room form the real meeting and the remote attendees become spectators. If more than two people are remote, run it fully remote — everyone on their own laptop, even those sitting in the same building. It feels absurd and it produces dramatically better participation.

Quantifying Business Value

Every engineering proposal competes for money against other proposals, and the one with a defensible number wins. The skill is not financial modelling — it is intellectual honesty under pressure to inflate. An FDE who presents a conservative, auditable, clearly-assumption-labelled value case earns credibility that compounds across every subsequent conversation; one who presents a heroic number gets it dismantled in public by a finance analyst.

The value hypothesis

TEXT

```
VALUE HYPOTHESIS — one page, written in week 1, revised continuously

WHO BENEFITS       Customer-service and order-management teams (11 FTE);
                   downstream: customers receiving on-time delivery
MECHANISM          Removing human queue time from credit-block exception
                   handling reduces order delay and agent effort
WE BELIEVE         Auto-releasing address-mismatch false positives removes
                   ~48% of exception volume from the human queue
WHICH DRIVES       (a) 390 agent hours/year returned
                   (b) ~1,900 fewer orders delayed by one day/year
WORTH              (a) EUR 23,400 at EUR 60/h fully loaded
                   (b) EUR 0 hard savings — but it is the thing the two
                       key accounts complained about in their QBRs
CONFIDENCE         (a) MEDIUM — rests on the 48% rate holding
                   (b) HIGH on count, UNQUANTIFIED on value
BIGGEST UNCERTAINTY The 48% auto-release rate, measured on 300 historical
                   cases from one month. Seasonal variation unknown.
HOW WE'D KNOW      Compare median response time and exception volume for
                   4 weeks post-go-live against the Jan 2026 baseline,
                   using the same query (Appendix B).
```

> **Say "we cannot value this" when it is true**
> The 1,900 fewer delayed deliveries probably matter more to the COO than the EUR 23,400, but you cannot honestly put a euro figure on them without a customer-churn model you do not have. Say exactly that. "The hard saving is EUR 23,400. The reason I think you should do this is the 1,900 deliveries, which I cannot price — but you can, and two of your accounts have already told you what they think it is worth." Refusing to invent a number is a credibility move, not a weakness.

Baselining — before you change anything

1. **Choose a metric that already exists in their systems.**
   A metric you invent will be disputed at measurement time. A metric from their own BI report will not.

2. **Write the query or the method down, in full, and share it.**
   Measuring "the same way" six months later is impossible unless the method is literally written out.

3. **Capture a long enough window to cover variation.**
   One month minimum; a full quarter if there is month-end or seasonal effect. A baseline taken in a quiet fortnight will make your results look like a miracle, and nobody will believe it.

4. **Record confounders at baseline time.**
   Headcount, volume, a system change, a pricing change, a public holiday pattern. In six months nobody will remember that three agents were on leave.

5. **Get the baseline agreed in writing by the person who will judge the result.**
   This is the single step that is always skipped and always regretted.

6. **Keep the raw extract.**
   Not the summary — the raw rows, in your own storage, with a date in the filename.

TEXT

```
BASELINE DEFINITION – Appendix B (shared, signed off by Priya 2026-02-07)

METRIC 1  First-response time
DEFINITION  For each inbound customer email, the elapsed wall-clock time
            between mail-server received_at and the first outbound reply
            from any CS agent to the same thread.
SOURCE      Exchange Online message trace joined to CRM activity table on
            internet_message_id.
EXCLUSIONS  Auto-replies; internal mail; spam-quarantined; threads with no
            reply within 14 days (counted separately as "unanswered").
WINDOW      1–31 Jan 2026 (22 working days)
N           6,310 messages
RESULT      median 26.1 h | p25 7.4 h | p75 41.0 h | p90 58.3 h
            unanswered after 14 days: 112 (1.8%)

METRIC 2  Credit-block exceptions requiring human action
SOURCE      SAP table VBUK credit status change log
WINDOW      1–31 Jan 2026
RESULT      mean 24.8/working day; Mondays mean 58.3/day; total 546

CONFOUNDERS RECORDED AT BASELINE
  – 9 CS FTE, no vacancies, 1 on long-term leave from 12 Jan
  – Inbound volume +11% vs Jan 2025
  – No system changes in the window
  – 1 public holiday (1 Jan, excluded from working days)

RE-MEASUREMENT  Identical query, 4-week window beginning 2 weeks after
                go-live (to exclude hypercare distortion).
AGREED BY       P. Menon, 2026-02-07, email ref NSD-0207-01
```

ROI, payback, NPV and cost of delay

TEXT

```
THE FOUR NUMBERS, WORKED

INVESTMENT
  Build (80 days @ EUR 1,200)                          96,000
  Customer internal effort (12 days @ EUR 450)          5,400
  Annual run cost (hosting, monitoring, 4 d/yr maint) 9,600
  ──────────────────────────────────────────────────────────
  Year 0 outlay                                       101,400
  Annual ongoing cost                                   9,600

ANNUAL BENEFIT (hard, conservative)
  390 agent hours returned @ EUR 60 fully loaded       23,400
  Reduced duplicate-handling of re-sent orders
    (est. 340 cases @ 12 min @ EUR 60)                  4,080
  ──────────────────────────────────────────────────────────
  Gross annual benefit                                  27,480
  Less annual run cost                                ( 9,600)
  NET ANNUAL BENEFIT                                    17,880

PAYBACK PERIOD
  101,400 / 17,880 = 5.7 years        ⟵ THIS DOES NOT PASS

HONEST CONCLUSION: on hard cost savings alone, this project does not clear
a typical 2-year payback hurdle. Do not hide that. Present it, then present
the case that actually justifies the spend:

THE REAL CASE
  - EUR 2.1m of renewal value at two accounts that cited responsiveness
  - ~1,900 delayed deliveries/yr removed
  - Elimination of a single-person dependency that currently has no cover
  - Monday backlog removal; 2 resignations in 18 months cited it
  Framed as: "EUR 101k to remove a named operational risk and address the
  issue two key accounts raised, with a EUR 17.9k/yr efficiency offset."

NPV (if you are asked, 3 years, 8% discount)
  Y0  -101,400
  Y1   +17,880 / 1.08      = +16,556
  Y2   +17,880 / 1.08^2    = +15,329
  Y3   +17,880 / 1.08^3    = +14,194
  NPV  = -55,321          ⟵ negative. Say so.

COST OF DELAY (the number that usually moves people)
  Net benefit per month = 17,880 / 12 = EUR 1,490
  Delayed deliveries per month        = ~158
  Renewal decision date               = 30 September
  >> Every month of delay costs EUR 1,490 and 158 delayed deliveries, and
     the window to show improvement before the September renewal closes
     at the end of June.
```

Present the negative NPV yourself

If the hard-dollar case does not work, say so before finance does. You will be in the room when it is discovered either way; the only variable is whether you are the person who found it or the person who hid it. An FDE who says "this does not pass on efficiency alone, here is why I still think you should do it" is trusted for years. One whose business case is picked apart by a controller is never asked for a number again.

Counting hours saved honestly

Dishonest move	Why it gets caught	Honest version
Minutes saved × headcount × 220 days	Implies 12 people become 11; they do not	Report hours returned; state explicitly that no headcount reduction is assumed
Using self-reported task times	Off by 2–3×; a sceptic will re-time it	Measure, or use system timestamps; state the method
Assuming 100% adoption from day one	Never happens; ramp is 2–3 months	Model a ramp: 40% / 70% / 90% over months 1–3
Counting the same saving in two projects	Finance consolidates and finds the double-count	Check the benefits register for overlapping claims first
Loaded rate = salary	Understates by ~35%; or overstates if you use a consulting rate	Ask finance for the fully-loaded internal rate; use their number
Ignoring the run cost	The thing you built must be hosted, patched and supported	Always subtract an explicit annual run cost
Claiming revenue you cannot attribute	Attribution is contested and you will lose	Present revenue at risk as context, not as benefit

The sentence that protects an efficiency case

"This returns roughly 390 hours a year to the team. I am not claiming a headcount reduction — that is your decision, not an outcome of this project. What I am claiming is that the Monday backlog disappears and that Martin stops being a single point of failure." Finance will test any claimed FTE saving against the actual payroll in twelve months. Hours returned is defensible; FTE saved usually is not.

The value model spreadsheet structure

TEXT

```
ONE WORKBOOK, FIVE TABS, NO HARDCODED NUMBERS IN FORMULAS

TAB 1  README
       What this models, who owns it, version, date, and a change log.
       The three sentences a reader needs before trusting any cell.

TAB 2  ASSUMPTIONS            ← the ONLY tab with typed numbers
       Every input on its own row, with:
         value | unit | source | confidence (H/M/L) | owner | last verified
       Example rows:
         exception_volume_per_day  24.8  count  SAP VBUK log Jan26  H  Arun 07Feb
         auto_release_rate         0.48  ratio  300 labelled cases  M  Arun 11Feb
         handle_time_minutes       4.5   min    observed n=19       M  Arun 06Feb
         loaded_hourly_rate        60    EUR    Finance (K. Weber)  H  Arun 05Feb
         adoption_month_1          0.40  ratio  assumption          L  -
       >> If a number is typed anywhere else in the workbook, it is a bug.

TAB 3  CALCULATION
       Pure formulas referencing Assumptions!. Laid out as a visible chain
       so a reviewer can follow it without clicking cells:
         exceptions/yr = volume_per_day * working_days
         auto_released = exceptions/yr * auto_release_rate
         hours_saved   = auto_released * handle_time_minutes / 60
         gross_benefit = hours_saved * loaded_hourly_rate
         net_benefit   = gross_benefit - annual_run_cost

TAB 4  SENSITIVITY
       Two-way table on the two lowest-confidence assumptions.
                     auto_release_rate
       handle_time   0.30    0.40    0.48    0.60
         3.0 min    9,190  12,254  14,705  18,381
         4.5 min   13,786  18,381  22,057  27,572
         6.0 min   18,381  24,508  29,410  36,763
       >> Present this table, not the single number. It pre-empts the
          "what if you're wrong about X" question by answering it.

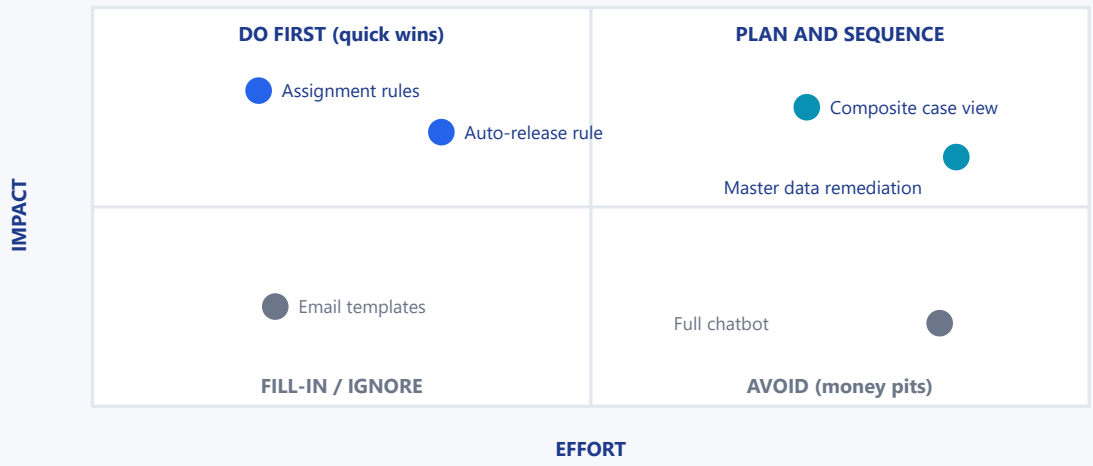
TAB 5  TRACKING
       Actuals, month by month, against the model. Populated from go-live.
       This tab is what turns a business case into a credibility asset:
       in month 4 you can show predicted vs actual and be right, or be
       wrong and explain why. Either is better than silence.
```

Prioritisation Frameworks

Prioritisation frameworks are not calculators that produce an answer; they are devices for making a group's implicit disagreements explicit so they can be argued about openly. The number a framework produces matters far less than the conversation it forces. Choose the lightest framework that will survive the political weight of the decision being made.

Impact versus effort

The default, and usually correct, choice. Two axes, four quadrants, ten minutes. Its weakness is that both axes are guesses, which is why it should be used for ordering within a workshop, not for a board-level investment case.



The chatbot the sponsor originally asked for sits bottom-right: high effort, low measured impact. Plotting it beside the alternatives makes that argument without you having to make it.

RICE

TEXT

```
RICE = (Reach x Impact x Confidence) / Effort

Reach       how many units (people, cases, orders) affected per period
Impact      3 massive, 2 high, 1 medium, 0.5 low, 0.25 minimal
Confidence  1.0 high (measured), 0.8 medium, 0.5 low (guess)
Effort      person-weeks

ITEM                        Reach   Imp  Conf  Eff   RICE
Auto-release rule           6,000    2   0.8   4.0   2,400
Assignment rules            6,310    1   1.0   1.0   6,310
Composite case view         6,310    2   0.5   6.0   1,052
Email templates (top 20)    4,200    1   0.8   2.0   1,680
Returns automation          1,120    3   0.5   8.0     210
Full chatbot                6,310    2   0.25 16.0     197

READ: assignment rules win decisively, because they are cheap, certain
and affect everything. Note this was the LOWEST-prestige item on the
original wish list and would have been built last.

WHERE RICE MISLEADS
 - Reach dominates the formula. A 10x reach difference swamps everything,
   which is wrong when a small group has disproportionate value.
 - Confidence is where people cheat. Enforce the rule: 1.0 requires
   measured evidence, and you must be able to name the measurement.
 - Effort as a single number hides risk. Keep the P/O spread visible
   alongside it.
 - It says nothing about dependencies or sequencing.
```

MoSCoW

Band	Means	Discipline
Must	Without this, the release has no value and should not ship	Must is capped at 60% of effort. If everything is Must, nothing is.
Should	Important, painful to omit, but there is a workaround	Name the workaround explicitly for each one, in writing
Could	Desirable; include if the time appears	These are your buffer. Say so openly.
Won't (this time)	Explicitly excluded from this release, with a date to revisit	The most valuable band. Write the revisit date next to every item.

The MoSCoW test that stops inflation

For each "Must", ask: "if we shipped on time without this, would you cancel the go-live?" If the answer is anything softer than yes, it is a Should. Ask it item by item, in the room, with the sponsor present. A list of eleven Musts typically shrinks to four in fifteen minutes.

Weighted scoring

TEXT

```
WEIGHTED SCORING - for when a group must own the trade-off explicitly

STEP 1: agree the criteria and weights BEFORE scoring anything.
        Weights must sum to 100. Agree them with no options visible.
        (If options are visible, people reverse-engineer weights to
        produce their preferred answer. This happens every time.)

  Criterion                        Weight  Agreed by
  Customer impact                    35    Priya, Rachel
  Speed to value (live in <8 wks)    25    Rachel
  Implementation risk (low=high sc)  20    Thomas
  Internal effort saved              15    Priya
  Reusability across other teams      5    Sanjay

STEP 2: score each option 1-5 on each criterion. Score INDIVIDUALLY and
        silently, then reveal. Discuss only the divergences.

  Option                 CI(35) SV(25) RI(20) ES(15) RU(5)  TOTAL
  Assignment rules         3      5      5      3     2      380
  Auto-release rule        4      4      3      5     4      395
  Composite case view      4      2      3      4     2      320
  Returns automation       5      1      2      4     3      315
  Full chatbot             4      1      1      3     5      255

  (TOTAL = sum of score x weight)

STEP 3: sanity check. "Does this ranking feel wrong to anyone?"
        If the model disagrees with the room's instinct, the model is
        usually missing a criterion. Find it, add it, re-run. Do NOT
        quietly adjust scores to reach a predetermined answer - that
        destroys the entire value of the exercise.
```

The "one thing by Friday" test

The most useful prioritisation question ever devised, and it fits on one line: "*If we could only ship one thing, and it had to be live a week on Friday, which one?*"

It works because it collapses three dimensions at once — value, feasibility and urgency — and because the artificial deadline prevents the comfortable answer. People who will happily rank ten items politely will suddenly reveal a strong preference when forced to pick one. Ask it after any formal scoring exercise. When the answer differs from the model's ranking, the gap is the most informative thing in the session, and you should spend five minutes on it.

Running a prioritisation session

TEXT

```
PRIORITISATION SESSION - 75 MINUTES

BEFORE
  - Candidate list capped at 12 items. More than 12 and the session fails.
  - Each item on one line: what it is, who benefits, rough size (T-shirt).
  - Send the list 48h ahead. No scores, no recommendation - just the list.
  - Decide and announce who DECIDES if the group cannot agree. Name them.

00:00  Restate the goal and the constraint                  [5 min]
       "We have 8 weeks and one engineer. We're deciding what fits."
       The constraint is what makes this real. Without a stated capacity
       limit, groups will rank items and then ask for all of them.

00:05  Walk the list - clarify only, no advocacy            [15 min]
       90 seconds per item. "Any questions on what this IS?"
       Interrupt advocacy firmly: "Hold that - we're clarifying now,
       arguing at 00:35."

00:20  Silent individual scoring                            [10 min]
       Each person scores all items on the agreed criteria, alone.

00:30  Reveal simultaneously                                [5 min]
       Put all scores on screen. Do not average yet.

00:35  Discuss ONLY divergence                              [20 min]
       "Everyone scored this 4 or 5 except Thomas, who scored it 1.
        Thomas, what do you know that we don't?"
       >> This is the entire point of the session. Thomas knows the
          composite view needs a firewall change with a 6-week lead time.
          No amount of averaging would have surfaced that.

00:55  Agree the cut line                                   [10 min]
       Draw a literal line. "These four fit in 8 weeks. These eight
       do not." Get verbal agreement from the decider, by name.

01:05  Name owners and the revisit date                     [10 min]
       Below-the-line items get a revisit date, not a silent death.

01:15  End. Write-up within 24 hours with the cut line in it.
```

⚠ **Never prioritise without a stated capacity constraint**

A group asked "what is most important?" will produce a ranked list and then request all of it. A group asked "what fits in eight weeks with one engineer?" makes real trade-offs. Always state the constraint first, in numbers, and repeat it whenever the conversation drifts toward "can we also...".

Managing Scope Creep and Saying No Well

Scope creep is rarely a single dramatic demand; it is forty small, reasonable requests, each of which costs half a day, each of which you grant to be helpful, and which together consume a month that you will later be blamed for losing. The defence is not rigidity — it is a mechanism that makes the cost of every addition visible at the moment it is requested, so that the customer, not you, decides what to give up.

The anatomy of creep

Form	Sounds like	Real cost	Response
The tiny ask	"While you're in there, could you also..."	Half a day each, forty times	Log it, size it, show the running total
Requirement drift	"When we said daily we meant hourly"	Architecture change	"That's a different requirement — here's the delta"
Gold plating (yours)	"I'll just make it configurable"	Weeks, invisibly	Self-discipline; YAGNI is a scope control
New stakeholder	"Legal has some additional requirements"	Can be enormous	Legitimate; raise as a formal change immediately
Scope by demo	"That's great — can it also do X?"	Unbounded	"Good idea — adding it to the backlog. Where would you rank it against Y?"
The quiet expansion	Two more regions appear in a test plan	Multiplies everything	Re-read the SOW out loud; raise a CR
Support creep	"Can you just look at this unrelated issue?"	Days per month	Time-box it, and report the time in status

The change request mechanism

TEXT

CHANGE REQUEST

CR-007 Raised: 2026-03-04
Raised by: Priya Menon
Project: NSD-2026-014 CS Response Time Phase 1

REQUEST
Include the Austrian subsidiary (NSD-AT) in phase 1, using the same auto-release rule.

REASON GIVEN
AT has the same problem and the AT country manager has asked to be included now rather than wait for phase 2.

ASSESSMENT
Technical: AT uses a separate SAP client (300 vs 100) with a different credit-block reason code set. The rule logic transfers; the configuration, mapping and testing do not.
Effort: +6 days (2 config, 2 mapping of AT reason codes, 1 test data, 1 UAT support)
Schedule: Go-live moves from 28 Mar to 09 Apr (+8 working days, including the AT works council notice period).
Cost: +EUR 7,200 against the EUR 96,000 cap. New cap EUR 103,200.
Risk: AT works council notification is a 10-working-day process that has not started. This is on the critical path and is not within our control.
Dependencies: AT SAP client read access (new request to Finance IT).

OPTIONS
A) Approve as above: +EUR 7,200, go-live 09 Apr for both DE and AT.
B) Reject: AT included in phase 2 as originally planned (May).
C) Partial: build the AT reason-code mapping now (+2 days, no schedule impact) so that phase 2 for AT becomes a 3-day deployment rather than a 6-day project. Go-live for DE stays 28 Mar.

RECOMMENDATION
Option C. It captures most of the benefit of including AT, costs 2 days instead of 6, and protects the DE go-live date which is tied to the September renewal conversation.

DECISION REQUIRED BY 11 Mar (after this date, option A is no longer available without a larger slip)

DECISION: [ ] A [ ] B [ ] C
Signed: Date:

✓ **Why the three-option CR works**

A change request with one option is a bill. A change request with three options, a recommendation, and a decision deadline is consulting. The customer retains control, sees the real trade-off, and almost always picks your recommendation — because you have shown your working. Option C in the example above is the move that makes you valuable: you found a cheaper way to give them most of what they wanted.

Five hard conversations, scripted

1. REFUSING ADDITIONAL SCOPE

TEXT

```
SITUATION: In a working group, the sponsor asks you to "also handle the
returns queue, it's basically the same thing."

WRONG
  "That's out of scope."                [true, hostile, ends collaboration]
  "Sure, I'll see what I can do."       [creep; you have just lost 8 days]
  "That's not really the same thing."   [argumentative, defensive]

RIGHT
  "Good idea, and I think you're right that it's valuable - returns was
   second on our list from the February workshop.

   It isn't the same mechanism though: credit blocks are a deterministic
   address comparison, returns need a judgement about condition and
   warranty status. I'd estimate 8 days rather than the 1 you might
   expect.

   Here's the trade: we have 8 weeks and 4 of them are committed. I can
   do returns instead of the composite case view, or we add it as a
   change request with 2 extra weeks and EUR 9,600, or we hold it for
   phase 2 which was the original plan.

   Which do you want? I'm genuinely happy with any of the three - it's
   your budget and your priorities."

WHY IT WORKS
 - Validates the idea (they are not wrong, and it was already on the list)
 - Supplies the information they lacked (it is not the same mechanism)
 - Converts "yes/no" into "which trade-off", which is the real question
 - Hands the decision back explicitly, which is where it belongs
 - You have not said no. You have made the cost visible.
```

2. THE DEADLINE THAT CANNOT BE MET

TEXT

```
SITUATION: You are three weeks in. Security review slipped two weeks.
The 28 March go-live is not achievable. Nobody has asked yet.

WRONG
  Saying nothing and hoping to recover.      [guaranteed worst outcome]
  Announcing it in the steering committee.   [ambush; sponsor humiliated]
  "We might be slightly delayed."            [vague; forces them to chase]

RIGHT - a phone call to the sponsor, before anyone else, same day:

  "Priya, I need to tell you something early rather than late. The
   28 March go-live is no longer achievable and I want you to hear it
   from me today.

   What happened: the security board moved our slot from 25 Feb to
   11 Mar. That's a 2-week shift on a gate that everything else sits
   behind. It isn't anyone's fault and I should have submitted a week
   earlier than I did.

   Where that leaves us: 11 April for the full scope.

   What I can do about it. Three options:
   1. Full scope, 11 April. Clean, tested, nothing cut.
   2. Go live 1 April with the auto-release rule and assignment rules
      only - that's 80% of the benefit - and the composite view follows
      in late April as a separate release.
   3. Keep 28 March by dropping the composite view entirely from
      phase 1. We would need to decide that this week.

   I recommend 2. It protects the measurement window you need before
   the September renewal conversation, and the composite view is the
   piece with least dependency on anyone else.

   What do you want to do, and who else needs to hear this from you
   before Thursday's steering?"

WHY IT WORKS
 - Earliest possible notification; sponsor is never surprised in public
 - States the cause factually, including your own part in it
 - Arrives with options, not a problem
 - Recommends one and gives the reasoning in their terms (September)
 - Final question protects the sponsor politically, which they will
   remember for the rest of the engagement
```

3. THE BAD IDEA, PROPOSED BY SOMEONE SENIOR

TEXT

```
SITUATION: The COO, in a steering committee, says the team should build
a customer-facing chatbot on top of the exception queue "to deflect
tickets", which would be slow, risky and solve a 15% problem.

WRONG
  "That won't work."                    [public contradiction of the buyer]
  "Sure, we can look at it."            [agreeing to something you will
                                         later have to unpick]
  A long technical explanation.         [nobody is listening by sentence 3]

RIGHT
  "Let me make sure I've got the goal: you want fewer customers needing
   to contact us at all. Is that right?"
                                        [reframe to intent, not solution]
  "That's worth going after. Can I give you one number before we decide
   how? Of the 26 hours a customer waits, 19 is queueing and 4 is the
   agent handling it. A chatbot works on the 4. Routing works on the 19.

   What I'd suggest: let me come back on Thursday with two sized options
   - a deflection approach and a routing approach - with the expected
   effect on that 26 hours for each. Then you can pick with the numbers
   in front of you rather than on my opinion in a meeting.

   If the deflection numbers are better I'll build it."

WHY IT WORKS
 - Never contradicts in public; converts opinion into a measurable question
 - Buys 48 hours, which is almost always enough
 - Offers to be proved wrong, which is disarming and occasionally happens
 - Gives the senior person a way to change their mind without losing face:
   they are not retreating, they are choosing between two options you
   brought them
```

4. YOU MISSED A COMMITMENT

TEXT

```
SITUATION: You promised the labelled dataset by Tuesday. It is Tuesday
afternoon and it is not done.

WRONG
  Silence until Wednesday.
  "Nearly done, just finishing up."         [if untrue, this is fatal]
  A paragraph of explanation.               [reads as excuse-making]

RIGHT — Tuesday morning, not Tuesday evening:

  "Priya — I said the labelled dataset today and it won't be ready until
   Thursday morning. The export only gave me 180 of the 300 cases because
   the archive job had already moved January; I've requested the archive
   restore and that takes 48 hours.

   Impact: none on the overall timeline — Thursday still leaves 4 days
   before the design review.

   What I'm doing: restore requested at 09:00 this morning, and I'm
   labelling the 180 I do have in the meantime so Thursday is just the
   remainder.

   I'll confirm Thursday 10:00."

WHY IT WORKS
 - Before the deadline, not after
 - One sentence of cause; no defensiveness, no blame
 - States the impact explicitly — usually it is small, and saying so
   prevents the customer imagining it is large
 - States the recovery action and a new specific time
 - Then: HIT THE NEW DATE. A missed recovery date is worth ten missed
   original dates.
```

5. ESCALATION

TEXT

SITUATION: Finance IT has not exposed the block_reason field. You asked on 6 Feb, chased on 14 Feb and 20 Feb. It is 25 Feb and it is now on the critical path. You must escalate without making an enemy of Anita.

STEP 1 – TELL ANITA FIRST. Always. Never escalate over someone silently.
"Anita, I know you're stretched and this isn't your priority – that's fair. But it's now holding up our go-live date, so I have to raise it at Thursday's steering as a blocker. I wanted you to hear that from me first rather than see it on a slide.
I'm going to describe it as a capacity issue, not a responsiveness issue, because that's what it is. Is there anything I should add about what you're carrying, so the ask lands on the right desk?"

STEP 2 – THE ESCALATION ITSELF. Facts, impact, specific ask, no blame.
"Blocker for decision: we need 2 days of Finance IT effort to expose one field (block_reason) on the credit-check output.

Requested 6 Feb through the standard ticket queue, currently 19 days old against a 10-day SLA. This is not a responsiveness problem on Anita's side – her queue has 40 open tickets and three P1s.

Impact: this sits on our critical path. Every week it slips moves go-live by a week. We are currently 2 weeks from losing the March date.

Ask: Rachel, can you get 2 days of Finance IT time allocated before 7 March? Alternatively we can write the change ourselves and have Finance IT review it, which needs 2 hours of Anita's time rather than 2 days – but that needs someone to agree we may touch their codebase."

STEP 3 – AFTERWARDS. Tell Anita what happened, same day, whatever the outcome. And when it is unblocked, credit her publicly in the next status report, by name.

WHY IT WORKS
- No surprise and no ambush; the relationship survives
- Defends the person while escalating the problem, which costs you nothing and buys an ally
- Offers a cheaper alternative (2 hours vs 2 days), which gives the executive an easy yes
- A specific ask with a date; "please help with Finance IT" achieves nothing

✦ **The universal shape of a hard conversation**

Fact, impact, options, recommendation, ask. Five beats, in that order, and it works for a slipped deadline, a rejected design, a budget overrun or a resignation. The two beats people omit are *impact* (so the listener invents their own, usually worse) and *ask* (so the conversation ends with nobody owning anything).

Expectation Management and Status Communication

Customers tolerate delay, cost and imperfection far better than they tolerate surprise. Almost every relationship breakdown an FDE experiences traces back to a moment when someone learned something late that you knew earlier. Status communication is therefore not reporting — it is the systematic prevention of surprise, and it is worth more of your attention than any single technical decision.

The no-surprises rule

BAD NEWS TRAVELS FIRST

The sponsor hears it from you, before the steering committee, before their boss, before the corridor. Same day you know it. This rule has no exceptions and no "I'll have better information tomorrow".

EARLY AND SMALL

A two-day slip reported immediately is a non-event. The same two days reported in week eight, as part of a four-week slip, is a crisis of confidence. Report the two days.

NEVER LET THEM BE EMBARRASSED

Your champion presenting your project to their peers must have current information. The fastest way to lose a champion is to let them say something in a meeting that you knew was no longer true.

The weekly status email

TEXT

```
Subject: NSD CS Response Time - Week 6 status - AMBER

TL;DR  Auto-release rule is working at 94% on test data. Security approval
       landed. We are AMBER because Finance IT capacity is still unresolved
       and it now has 4 days of float left before the March date moves.
       One decision needed from you by Friday.

_____________________________________________________________________

DECISION NEEDED FROM YOU
 1. Finance IT - 2 days of effort before 7 Mar, OR permission for us to
    write the change and have them review (2 hours of their time).
    Needed by: Friday 28 Feb.
_____________________________________________________________________

DONE THIS WEEK
 - Auto-release rule built; runs against 300 labelled historical cases
   with 94% agreement against the manual gold set, 0 false releases on
   the 40 true-block cases. (Target was 95%; see "watching" below.)
 - Security review board approved, conditional on a pen test before
   go-live. Pen test booked for 18 Mar.
 - Assignment rules configured in the CS tool; tested with 20 messages.
 - Martin reviewed the address normalisation rules and found two cases
   we had wrong (Austrian postcodes, and "c/o" lines). Both fixed.

NEXT WEEK
 - Composite case view: first working version, demo Thursday 15:00.
 - Re-run the rule against a fresh 300-case sample from December to
   check the 48% rate holds across months.
 - Draft runbook.

RAG: AMBER
 Why amber: the Finance IT dependency has consumed all but 4 days of its
 float. If it is not resolved by 7 Mar the 28 Mar go-live moves.
 What would make it green: a named Finance IT resource with a date.

RISKS AND ISSUES
 R1 Finance IT capacity        [OPEN, escalating Thursday]   Owner: Priya
 R2 94% vs 95% accuracy target [WATCHING]                    Owner: Arun
    The 6% disagreement is entirely cases where the human ALSO released
    the block, but for a different stated reason. Analysing whether the
    gold set is mislabelled rather than the rule being wrong. Will report
    Tuesday.
 R3 Pen test on 18 Mar leaves 6 working days to remediate findings
    before go-live. Tight but acceptable. [ACCEPTED]

BURN
 Consumed 31 of 80 days (39%). Elapsed 6 of 10 weeks (60%).
 Ahead on budget because discovery ran short. No action needed; I will
```

```
 flag again at 70%.

WHAT I NEED FROM THE TEAM
 - Martin: 2 hours Thursday for the composite view walkthrough
 - Sunita: confirm the 9 agents' availability for training w/c 24 Mar

Arun
```

> **Structure notes on that email**
> The decision request is above the fold and visually separated, because the sponsor may read only the first screen. The RAG status is justified with a specific trigger ("4 days of float") and a specific exit condition ("a named resource with a date") rather than a feeling. The accuracy shortfall is reported even though it is probably a labelling artefact — reporting a miss before you have the explanation is what makes the explanation believable when it arrives.

RAG honesty

Status	Means	Trigger	What must accompany it
Green	On track; no help needed	All milestones have float; no open blockers	Nothing
Amber	At risk; a decision or resource is needed	A dependency has consumed most of its float, or a risk has become likely	The specific thing that would return it to green
Red	A committed date or outcome will not be met	Float exhausted; scope, date or cost must change	Three options and a recommendation

> **Watermelon status**
> Green on the outside, red in the middle. It happens because amber feels like admitting failure and because an optimistic engineer believes they will recover by Friday. The cost is catastrophic: when the project finally turns red it goes from green to red in one week, and every previous green report is retroactively read as dishonesty. Rules that prevent it: amber must be used at least once in any multi-month project (if you were never at risk, you were not being honest); any status change must be reported within 24 hours, not saved for the weekly; and if you are debating between green and amber, it is amber.

The escalation ladder

TEXT

```
DEFINE THIS IN WEEK 1, WRITE IT DOWN, AGREE IT WITH THE SPONSOR

LEVEL 0  Direct, same day
         I raise it with the person who owns it. Most things end here.
         Record: a line in the open-questions list.

LEVEL 1  Working group, weekly
         Unresolved after one chase, or affects more than one team.
         Record: action with owner and date.

LEVEL 2  Sponsor, as it happens
         On the critical path, or needs a resource the working group
         cannot commit. I call Priya. Not an email.
         Record: decision log entry.

LEVEL 3  Steering committee, monthly or ad hoc
         Needs budget, cross-functional authority, or a scope change.
         Record: decision log entry with the named decision-maker.

LEVEL 4  Commercial escalation (my account lead to their procurement)
         Contractual dispute, payment, or relationship breakdown.
         Rare. If you reach here, the relationship has already failed.

TIMING RULE  Nothing sits at one level for more than one cycle. If it is
             unresolved after a week at Level 1, it goes to Level 2
             automatically, and I tell the owner before it moves.

NAMING RULE  Escalate the problem, never the person. "This dependency is
             unresolved and here is what it costs" - not "Anita hasn't
             done it."
```

Delivering bad news

1. **Within hours, not days.**
   The decay rate on bad news is brutal: the same fact, delivered a week later, reads as concealment rather than reporting.

2. **Verbally first, in writing second.**
   Call, then send the written version within the hour so there is a record. Never deliver significant bad news by email alone.

3. **Lead with it.**
   No warm-up, no three paragraphs of context. "I need to tell you that the March date has moved." Burying it is read as manipulation.

4. **One sentence of cause, no defence.**
   Resist the urge to establish that it was not your fault. If some of it was your fault, say that plainly and once; it costs you nothing and buys enormous credibility.

5. **State the impact precisely.**
   Unquantified bad news gets catastrophised. "Two weeks, no cost impact" is a much smaller event than "there's a problem with the timeline".

6. **Arrive with options.**
   Never deliver a problem without at least two paths and a recommendation.

7. **Say what you have already done.**
   "I've already booked the alternative slot" converts you from the messenger into the person handling it.

8. **Ask who else needs to know, and offer to help them tell those people.**
   You are protecting your sponsor's position, not just reporting a fact.

✓ **The counter-intuitive result**
Engineers who report problems early and precisely are trusted more than engineers whose projects never visibly have problems — because everyone knows every project has problems, so an unbroken run of green reports reads as either luck or concealment. Visible, well-handled bad news is a demonstration of competence. It is also the only reliable way to be given a harder engagement next time.

Handling Conflict and Difficult Stakeholders

Almost nobody is difficult without a reason, and the reason is almost never the one they state. Behind resistance there is usually a rational interest — a threatened budget, an unacknowledged workload, a previous failure they were blamed for, a genuine technical objection nobody has listened to. Diagnosis before response is the whole discipline. The engineer's instinct, which is to out-argue the objection, is the one response that reliably fails.

Diagnosis first

TEXT

```
THE DIAGNOSTIC QUESTIONS — ask yourself before you act

1. What does this person lose if I succeed?
   Budget, headcount, status, a system they own, a narrative they have
   been telling, a programme they are running, their indispensability.

2. What have they been burned by before?
   Ask explicitly: "has anything like this been tried here before?"
   The answer reframes most resistance as pattern-matching, not hostility.

3. Are they objecting to the WHAT or the HOW I ARRIVED?
   Very often the objection is procedural — they were not consulted — and
   the content is fine. This is cheap to fix and expensive to misdiagnose.

4. Is their objection actually correct?
   Sit with this one honestly for a full minute. Hostile stakeholders are
   right surprisingly often; they are just rude about it.

5. Do they have the power to stop me, or only to slow me?
   This determines whether you must convert them or merely route around
   them. Spending three weeks converting someone who cannot stop you is
   a waste; routing around someone who can is fatal.

6. What would a win look like for them?
   If you cannot answer this, you have no strategy, only an argument.
```

The resistant SME

TEXT

```
PRESENTATION
Knows the domain better than anyone. Answers questions in monosyllables.
Says "it's more complicated than that" without elaborating. Does not attend
sessions. Finds a fatal flaw in everything you propose, late.

LIKELY CAUSE
 - Their expertise IS their job security, and you are proposing to encode it
 - They have watched three consultants simplify their work and get it wrong
 - Nobody has asked their opinion before deciding, only after
 - They are genuinely right that it is more complicated

RESPONSE
 1. Make them the authority, visibly and in writing. Name them as process
    SME in the SOW. Say "this rule is Martin's, I just implemented it."
 2. Ask them to find the flaws, formally. "I need you to break this. What
    are the cases where this rule gives the wrong answer?" Being asked to
    criticise is the role they want and are good at.
 3. Show them everything FIRST, before any other stakeholder. The status
    of being the first reviewer is worth more than any argument.
 4. Never present a simplification of their work to management without
    them in the room.

WORDS THAT WORK
 "You've been doing this eleven years and I've been here three weeks.
  I'm going to get things wrong and I need you to tell me when I do -
  ideally before I show it to Priya rather than after."

 "I've written down the rule as I understood it. I'm fairly sure it's
  wrong in at least two places. Where?"

WHAT NOT TO DO
 Do not go around them to a more cooperative junior person. The SME will
 find out, will be right that you did it, and will stop cooperating
 permanently. Their knowledge is not replaceable.
```

The absent sponsor

TEXT

PRESENTATION
Enthusiastic at kickoff. Cancels two of the next three 1:1s. Replies to emails in four words after three days. Does not attend steering. Decisions wait weeks.

LIKELY CAUSE
- Reorganisation, a bigger fire, or a changed personal priority
- The project was someone else's idea and they inherited it
- They do not actually control the budget and are avoiding saying so
- You are asking for too much of their time, or asking badly

RESPONSE
1. Reduce the cost of engaging with you. Replace the 30-minute meeting with a 5-line email requiring a one-word answer. Make decisions binary and deadline them with a default.
2. Use the default-decision mechanism explicitly (see below).
3. Find the deputy. Ask directly: "when you're not available, who can make this call?" Get that in writing.
4. Diagnose the real cause, directly and without accusation, in person.
5. If there is genuinely no sponsor attention available, escalate the structural issue once, in writing, and then manage your own exposure.

THE DEFAULT-DECISION EMAIL - the single most useful tool here
  Subject: Decision needed - address normalisation standard - default applies Friday

  Priya - one decision, 30 seconds.

  Should "Strasse" and "Str." be treated as the same address?

  A) Yes - normalise abbreviations before comparing (my recommendation; affects ~14% of cases, slightly higher auto-release rate)
  B) No - exact match only (safer, fewer auto-releases)

  Unless you tell me otherwise by Friday 17:00, I will proceed with A and record it in the decision log as a default decision.

  Arun

>> This is not passive aggression. It is a legitimate mechanism, it must be agreed as a working practice up front, and the default must always be the lower-risk or the reversible option. It moves projects that would otherwise stall for months.

WHEN TO STOP
If a project has had no sponsor engagement for six weeks, write one clear email stating that the project cannot progress without decisions X, Y and Z, state the cost of continuing without them, and propose pausing. A

```
paused project you can restart is better than a slow-motion failure you
will be associated with.
```

The hostile IT team

TEXT

```
PRESENTATION
Access requests sit for weeks. "That's not how we do things here."
Architecture review raises new requirements each time. You hear "we weren't
consulted on this" in month three.

LIKELY CAUSE — usually one of these four, and they are all legitimate
 1. They were genuinely not consulted, and a business function hired you
    to route around them. They are correct to be annoyed.
 2. They will own support for whatever you build, and nobody asked them.
 3. They have real standards and have watched a dozen business-led
    projects dump unsupportable systems on them.
 4. Capacity. They are not hostile, they are drowning.

RESPONSE
 1. Go to them early, before you need anything, and ask for their
    standards. "What do I need to satisfy to get this through your
    review? Can I have the checklist?" Almost nobody does this and it
    changes the relationship immediately.
 2. Accept their standards wherever you possibly can. Each accepted
    standard buys you an exception somewhere you actually need one.
 3. Name their support model as a deliverable. "You'll own this from
    April. What do you need from me so that's not a burden?"
 4. Give them a seat in the working group and credit in the status report.
 5. Never win an architecture argument by escalating to a business
    sponsor. You will win the battle and spend the rest of the
    engagement in a queue.

WORDS THAT WORK
 "I'd rather find out what fails your review now than in March. Can I
  walk you through the design for twenty minutes and have you tell me
  what you'd reject?"

 "You're going to own this after I leave. I'd like to build it the way
  you'd want to support it. What does that look like?"
```

The political land-grab

TEXT

```
PRESENTATION
A peer leader asserts that your work belongs under their programme, their
CoE, their platform, their governance. Appears in your steering committee
uninvited. Proposes a "merge" that would make you a workstream.

LIKELY CAUSE
 Their programme needs visible wins, or your success makes their
 investment look slow, or they genuinely have a mandate you were not told
 about.

RESPONSE - in order of preference
 1. GIVE AWAY THE CREDIT, KEEP THE DELIVERY. This is almost always the
    right answer and engineers hate it. Offer co-branding, offer that the
    component becomes a reusable asset in their catalogue, offer them the
    steering committee slot to present it. You lose nothing you actually
    need.
 2. Find the genuine overlap and formalise it. If they have a rules
    engine, use it. Dependency is cheaper than war.
 3. Clarify the mandate upward, once, neutrally: "Rachel, Sanjay and I
    both think this sits in our area. Rather than negotiating it between
    us, can you tell us where you want it to sit?" Then accept the answer.
 4. Only if 1-3 fail: document the duplication and its cost, present it
    factually, and let the organisation decide. Never frame it as a
    personal dispute.

WHAT NEVER WORKS
 Competing on technical merit. Being right about architecture has no
 bearing on an organisational boundary dispute. You will produce an
 excellent comparison document that nobody reads and lose anyway.
```

The customer who wants a different product

TEXT

PRESENTATION

"Can it also do forecasting?" "Does it integrate with Workday?" "What we really need is a full CRM replacement." Each demo generates requests that are further from what you actually have.

LIKELY CAUSE

- They have a bigger problem than the one they hired you for
- Your product is the first thing in years that has worked, so it is being asked to carry everything
- Nobody has ever written down the boundaries of what they bought

RESPONSE

1. Separate the roadmap conversation from the delivery conversation. Two different meetings, two different documents. Mixing them turns every delivery review into a negotiation.
2. Maintain a visible "not in this product, but noted" list and review it formally once a month. People accept exclusion far better when they can see their item written down and dated.
3. Be honest fast about hard noes. "We will not be building a CRM. If that is what you need, I can tell you who does it well." Saying this in month one costs you nothing; saying it in month eight after letting them hope is a breach of trust.
4. Find the version of their request that IS in scope. "We can't do forecasting, but we can expose the exception data so your existing forecasting tool can consume it. Would that solve it?"

WORDS THAT WORK

"I want to separate two conversations. One is 'is what we're building working' - that's today. The other is 'where could this go' - that's worth an hour on its own and I'd like to schedule it properly rather than squeeze it into the end of a demo."

Conflict mechanics that generalise

Move	Why it works	Example phrasing
Separate people from the problem	Allows disagreement without a loser	"We both want the March date. We disagree on what it takes."
Find the interest behind the position	Positions conflict; interests often do not	"Why is on-prem important to you?" → it is auditability, which you can solve
Agree the facts before the decision	Most arguments are factual disputes in disguise	"Before we decide, can we agree how many cases this affects?"
Use the written record	Removes memory as a battleground	"Decision log D-003, 18 Feb, agreed by Rachel."
Steelman their case out loud	Proves you listened; often improves your own position	"The strongest version of your argument is X. Have I got it right?"
Offer the choice, not the answer	Preserves their authority	"Here are the two paths. You own this decision."
Concede early and visibly	Concessions on small things buy credibility on large ones	"You're right about the audit log. I'll change it."
Move it offline	Public disagreement hardens positions	"Let's take this out of the meeting — I'll come to you at four."

⚠ **The one conflict you must not avoid**

When someone is factually wrong in a way that will cause real damage — an unsafe design, a legally non-compliant data flow, a number in a board paper that is not true — you must say so, clearly, on the record, even at cost to the relationship. State it once, calmly, in writing, with the reasoning. If it is overruled, record that it was overruled and by whom. You are not obliged to win; you are obliged to have said it.

Influence Without Authority

An FDE has no line authority over anyone whose cooperation they need. Everything you achieve inside a customer organisation is achieved through influence, and influence is not charisma — it is an accumulation of mundane, repeatable behaviours: being reliably

useful, being easy to say yes to, pre-wiring decisions so that meetings ratify rather than debate, and making other people's success the mechanism of your own.

Credibility is built from small, kept promises

COMPETENCE

Demonstrated, not claimed. One working thing beats any credential. The fastest credibility move available to an FDE is to solve a small, visible, annoying problem in week one that nobody asked you to solve.

RELIABILITY

You do what you said, when you said. This is 80% of trust and it is entirely within your control. Track every commitment you make in a list, including the trivial ones made verbally in corridors.

INTIMACY

People tell you things. Earned by discretion: never repeat a confidence, never be the source of gossip, never use something told to you privately in a public argument.

LOW SELF-ORIENTATION

The strongest trust multiplier. Visibly optimising for their outcome rather than your invoice. Telling a customer not to buy something is the purest form of this.

SPECIFICITY

"About 48%, measured on 300 cases from January" is credible. "Roughly half" is not. Precision signals that you have actually done the work.

ADMITTING LIMITS

"I don't know, I'll find out by Tuesday" raises your credibility. Confident guessing destroys it the first time you are caught, and you will be caught.

TEXT

```
THE COMMITMENT LEDGER — keep one, review it every morning

DATE   TO       I SAID I WOULD                          BY      DONE
04 Feb Martin   Send my interview notes                05 Feb  Y
05 Feb Priya    Find out if CreditPro has an API        07 Feb  Y
06 Feb Anita    Send the exact field spec, 1 page       07 Feb  Y
06 Feb Sunita   Look at why the Monday report is late   11 Feb  Y
07 Feb Thomas   Share the design before the board       20 Feb  Y
11 Feb Martin   Show him the rule before anyone else    14 Feb  Y
12 Feb Klaus    Answer his 4 questions in writing       13 Feb  Y

RULE: corridor commitments count. "I'll look into that" said while
walking to the lift is a promise and will be remembered as one.
Either write it down and do it, or do not say it.

RULE: over-deliver on small things, never on large ones. Sending notes
a day early is free credibility. Secretly building extra features is
not over-delivery, it is scope creep you inflicted on yourself.
```

Reciprocity, used honestly

People help people who have helped them. This is not manipulation if the help is real and given without a ledger. The practical form for an FDE: spend a deliberate portion of your time on small acts of usefulness that are not on your workplan.

- [ ] Fix the broken report that the team lead has complained about for a year. It takes you two hours.
- [ ] Write the SQL that someone has been asking IT for since November.
- [ ] Send a relevant article, a tool, or an introduction with no ask attached.
- [ ] Credit someone by name in a status report that their manager reads.
- [ ] Take the note-taking job in a meeting you did not call.
- [ ] Explain something technical to a non-technical person patiently, when it is not your job.

The return is not transactional and should not be treated as such, but it is real: when you need a favour at short notice in week nine, the people who have experienced you as useful will find the time.

Framing

Instead of	Say	Because
"We need to refactor the ingestion layer"	"We can cut the overnight run from 4 hours to 40 minutes, which means the morning report is ready before the stand-up"	Frame in the listener's observable world, not yours
"This will save 390 hours"	"This gives Martin his Mondays back and means the queue doesn't stop when he's on leave"	Concrete and human beats aggregate and abstract
"We should adopt X"	"Two of your peers in the sector moved to X after the same problem"	Social proof carries more weight than argument in risk-averse organisations
"If we do this, we'll gain Y"	"If we don't do this before September, we go into the renewal with the same numbers they complained about"	Loss framing moves more decisions than gain framing
"Can we talk about priorities?"	"I need 20 minutes to decide between two options; I have a recommendation"	Bounded, specific asks get accepted; open ones get deferred
"The team thinks..."	"I recommend..."	Hiding behind a collective reads as evasion and weakens the ask

Pre-wiring decisions

No significant decision should be made for the first time in the meeting where it is formally taken. If a steering committee is the first time a stakeholder hears your recommendation, you have given them a public stage on which to object, and public objections harden into positions.

TEXT

```
PRE-WIRING SEQUENCE — for a decision at Thursday's steering committee

MON     Draft the recommendation. One page. Three options, a
        recommendation, a decision deadline.

MON pm  Champion first (Priya). 20 min, in person.
        "Here's what I'm going to recommend on Thursday. Before I finalise
         it — what's wrong with it, and will Rachel buy it?"
        >> Your champion's edits make it hers. She will then defend it.

TUE     The likely objector (Sanjay). Coffee, no agenda.
        "I'm recommending option B on Thursday. I wanted you to see it
         first because it touches your CoE. What would you change?"
        >> If he objects, you now have 48 hours to adjust or prepare, in
           private, instead of 30 seconds in public.

TUE     The technical authority (Thomas). 15 min.
        "Does anything here fail your standards? I'd rather fix it now."

WED     The quiet one who will be asked for an opinion (Sunita).
        Often the person whose silent nod the sponsor watches for.

WED pm  Send the pack. Nothing in it is new to anyone who matters.

THU     The meeting ratifies a decision that has already been made in
        four conversations. It takes eleven minutes.

COST: about two hours.
SAVING: the three weeks a deferred decision costs, and the political
capital lost in a public disagreement.
```

✦ **The pre-wiring question**

"What would make you say no to this?" asked privately, 48 hours before a decision meeting. It is the highest-yield question in organisational life. People will tell you, honestly, in private, the objection they would otherwise raise in public — and most such objections are cheap to address once you know them.

Coalition building

1. **Start with the person who already agrees.**
   Not the hardest to convince. Early, visible support makes later conversations easier because people assess positions socially, not just logically.

2. **Make the coalition visible.**

"Thomas has reviewed this and is comfortable; Martin has validated the rules" removes the fear of being the first to commit.

3. **Give each member a role.**
Reviewer, presenter, pilot user, data provider. People defend what they have contributed to far more strongly than what they merely approved.

4. **Let others present.**
When your champion presents your work as theirs, you have won. Resist the urge to be the one holding the clicker.

5. **Keep the coalition informed even when nothing is happening.**
A two-line update in a quiet month keeps people attached.

Getting a decision out of a slow organisation

Symptom	Underlying cause	Lever
"We need to discuss it internally"	Nobody knows who decides	Ask directly: "who makes this call?" Then go to them.
Decision deferred three meetings running	The decision feels large and irreversible	Shrink it. "Can we decide only the pilot scope today?"
Everyone agrees, nothing happens	No owner, no date	"Who owns this and by when?" out loud, in the room
Endless information requests	Risk aversion; nobody wants to be the one who approved it	Make it reversible and say so: "we can turn this off in ten minutes"
Decision made, then unmade	Someone was not consulted	Find them, consult them, re-take the decision properly
"Let's take it to the architecture board"	Genuine governance, or a polite kill	Ask the board's chair what a successful submission looks like
Silence	Competing priority elsewhere	Default decision with a deadline; reduce the cost of a yes

Shrink the decision

The most reliable technique for unsticking an enterprise is to make the decision smaller and reversible. "Approve the programme" is a career risk; "approve a four-week pilot with 9 users that we can switch off" is not. Get the small yes, deliver, and the large yes becomes a formality because the evidence now exists. Most big enterprise decisions are made retroactively, by ratifying something that is already working.

Closing an Engagement

The last two weeks determine what the engagement is remembered as. An FDE who builds something excellent and leaves it undocumented, unadopted and unowned has produced a liability for the customer and a reference they will not give. Closing well is a distinct skill with its own artefacts, and it must be planned from week one — not improvised when the budget runs out.

Adoption is the real deliverable

TEXT

```
ADOPTION PLAN - start writing this in week 2, not week 20

WHO MUST CHANGE THEIR BEHAVIOUR
  9 CS agents            must work from an assigned queue, not a shared inbox
  1 team lead (Sunita)   must monitor the SLA timer and rebalance
  Martin                 must stop manually clearing the queue (hardest)
  Finance IT (Anita)     must own the block_reason field going forward

WHAT COULD STOP THEM
  - Habit. The shared inbox has been the way for six years.
  - Distrust of the auto-release. "What if it releases a real block?"
  - Fear of measurement. SLA timers look like surveillance.
  - Martin's identity is partly built on being the person who clears it.

INTERVENTIONS
  W-6  Agents see weekly demos; two of them sit in the working group
  W-4  Publish the FAQ: "what this means for your job" - written plainly,
       addressing headcount directly and honestly
  W-3  Pilot with 3 volunteers, not a big-bang switch
  W-2  Martin co-presents the rule to the team. He explains it, not me.
  W-1  Training: 2 sessions, 90 min, hands-on, recorded
  W0   Go-live. I sit with the team all day. Not on a call - physically.
  W+1  Daily 10-min check-in with Sunita
  W+2  Weekly. Publish the first measured numbers to the whole team.
  W+4  Celebrate publicly, naming agents not the technology

MEASUREMENT OF ADOPTION (distinct from measuring the outcome)
  - % of exceptions handled through the assigned queue vs shared inbox
  - Number of manual overrides of the auto-release decision
  - Logins by distinct user per week
  - Unprompted questions from agents (silence is a bad sign, not a good one)
```

The silent failure

The system works, the metrics look fine, and six months later you discover the team maintained a parallel shadow spreadsheet the whole time because one edge case was never handled and nobody wanted to tell you. Watch for it: unexplained gaps between system volume and real volume, a persistent "miscellaneous" category, people who are always busy but never in the system. Ask directly and without judgement: "what are you still doing outside the tool?"

Train the trainer

1 **Pick the trainer early**

— ideally in week three — and make it someone respected by peers rather than the most technical person.

2. **They shadow you.**
   They sit in on design sessions and build sessions so they understand why, not only what.

3. **You deliver session one; they observe and take notes.**

4. **They deliver session two; you sit at the back and say nothing.**
   This is the hard part and the whole point.

5. **They answer the questions.**
   When an attendee looks at you, look at the trainer. If they get it wrong, correct it afterwards in private, never in the room.

6. **Hand over the materials in editable form**
   — not a PDF — with a note on what to update when things change.

7. **Be publicly available for two weeks and then stop.**
   Announce the end date so the transition is explicit rather than a slow fade.

The handover pack

TEXT

```
HANDOVER PACK - table of contents
(The technical contents are covered in depth in the Technical Writing
guide; this is the consulting-side checklist of what must exist and
who must sign that it does.)

1.  Executive summary - what was built, why, what it achieves    [1 page]
2.  Architecture overview with a diagram                          [2 pages]
3.  Environments: dev/test/prod, URLs, how to access
4.  Credentials and access inventory - WHO has what, how to grant
    and revoke, and what breaks if a named person leaves
5.  Operational runbooks - one per known failure mode
6.  Monitoring and alerting - what fires, to whom, what to do
7.  Known issues and accepted limitations, with ticket references
8.  Deferred items and technical debt register, with estimates
9.  Data: sources, schemas, retention, lineage, PII locations
10. Support model: who is first line, second line, escalation, hours
11. Contacts: every person involved, role, what they know
12. Change history and the decision log
13. How to make the three most likely future changes
14. Benefit measurement: the baseline, the query, the result

THE ACCEPTANCE TEST FOR THIS PACK
Not "is it written". The test is: a named operator who was not on the
project performs three scenarios using only the written material, with
you in the room but silent. Observe where they hesitate. Those are the
gaps. Fix them and repeat. Document the result in the acceptance note.

THE QUESTION TO ASK YOURSELF
"If I am unreachable for six weeks starting tomorrow, what breaks?"
Whatever you answer, that is what is missing from the pack.
```

Success metrics review

TEXT

```
CLOSING REVIEW - 60 MIN - sponsor, champion, key users, you

AGENDA
00:00  The baseline we agreed, read back verbatim                [5 min]
       "On 7 February we agreed: median 26.1h, measured this way."
00:05  The result, same query, same method                       [10 min]
       Median 5.2h over 4 weeks (3 Apr - 1 May). Exceptions requiring
       human action down 46%. Monday backlog: eliminated.
00:15  What we predicted vs what happened                        [10 min]
       Predicted 48% auto-release; actual 46%. Predicted 390 hours
       returned; actual run-rate implies 372. Honest variance analysis,
       including the two things we got wrong.
00:25  What did NOT work                                         [10 min]
       The composite case view is used by 2 of 9 agents. Why: agents
       keep SAP open anyway for other tasks, so the aggregation saves
       less than we modelled. Lesson recorded.
00:35  What remains - deferred items and their estimates         [10 min]
00:45  Support model from today, and who owns what               [10 min]
00:55  Thank you, by name, specifically                          [5 min]

THE DISCIPLINE: present the failure (00:25) with the same energy as the
success. A closing review that is entirely positive is read as marketing
and devalues the positive parts.
```

References, expansion and the exit conversation

TEXT

```
THE EXIT CONVERSATION – sponsor, 45 minutes, no deck, final week

Ask these and then be quiet:

 1. "What did we do that was most useful?"
 2. "What did we do that wasted your time?"
 3. "What would you tell someone considering working with us?"
 4. "Where did I personally frustrate you? I'd rather know."
 5. "What should we have done differently in the first month?"
 6. "Is there anything unresolved between us that I should know about?"

THEN, and only after listening properly:

 7. "Would you be willing to act as a reference?"
    - Be specific about what that means: a 20-minute call, twice a year,
      with prospects in a similar situation. Not a logo on a website
      unless you ask separately and in writing.

 8. "Is there anyone else in the business with a version of this problem?"
    - An internal introduction from a satisfied sponsor is worth more
      than any sales activity. Ask for the introduction explicitly;
      hinting does not work.

 9. "What would make you want us back?"
    - This answer tells you whether there is a phase 2, honestly,
      before anyone writes a proposal.

RULES
 - Ask about expansion AFTER the value is demonstrated and measured,
   never before. Asking in month two reads as what it is.
 - If the engagement went badly, ask questions 1–6 and stop. Do not ask
   for a reference. Send a short written summary of what you heard and
   what you will change, and leave it there. That note has produced more
   returning customers than any recovery attempt.
 - Leave the door open without pressure: "If something comes up, you have
   my number. I'd be glad to help and equally glad to tell you if we're
   not the right people for it."
```

✓ **The best possible ending**

The customer's own team demonstrates the system to another part of the business, without you present, and gets the details right. That is the only outcome that proves the capability actually transferred. Design the whole engagement backwards from that moment.

Consulting Ethics and Boundaries

An FDE sits inside a customer's organisation with access to their data, their failures and their internal politics, while being paid by someone else. That position creates obligations that no contract fully specifies. The reputational asset you are building is not technical — it is the belief that your assessment is honest even when honesty costs you revenue. It takes years to build and one engagement to destroy.

Honest assessment

THE PRESSURE

Your pipeline needs a phase 2. The sponsor wants to hear that the pilot succeeded. Your own account team has a number. Every force in the system pushes toward a more favourable reading of ambiguous results.

THE PRACTICE

Report the measurement you agreed, by the method you agreed, before you look at whether you like it. Pre-register the method in writing. If the result is poor, present it with the same structure and confidence as a good result.

TEXT

```
WHAT AN HONEST NEGATIVE RESULT SOUNDS LIKE

"The pilot did not achieve its target. We said median response below 6
hours; we got 9.4.

Why: the auto-release rate held at 46% as predicted, but agents are not
using the assigned queue - 61% of exceptions are still being picked up
from the shared inbox, which reintroduces the queueing we were trying
to remove. That is an adoption problem, not a technical one, and it is
partly mine: I under-invested in the change side and over-invested in
the rule.

What I think now: the remaining 3.4 hours is addressable, but by
management attention rather than engineering. Sunita turning off the
shared inbox would probably do more than anything I could build.

What I would do: give it four weeks with the inbox closed and re-measure
before spending any more money with us. If it works, you need no further
engineering. If it doesn't, call me and I'll look at it again."
```

Recommending less work than you could sell

"Give it four weeks before spending more money with us" is the single most valuable sentence in that script. It demonstrably costs you revenue, which is precisely why it is believed — and it is why the same customer calls you first next time, with a bigger problem and no competitive process.

Not overselling

Overselling	Honest version
"This will transform your customer service"	"This addresses 19 of the 26 hours. The remaining 7 is agent handling time, which this does not touch."
"Our platform handles that"	"It handles the common case. Your Austrian subsidiary's reason codes would need mapping — about two days."
"We've done this many times"	"We've done three similar engagements. None of them had your SAP client split, so I'd treat that as a genuine unknown."
"AI can read those documents"	"On scanned documents of this quality, expect 80–90% field accuracy. You will need a human review step for anything financial."
"We can start Monday"	"We can start Monday but access typically takes three weeks here, so useful work begins around the 20th."
"It's basically done"	"The happy path works. Error handling, the audit log and the deployment are not started — about two weeks."

Confidentiality in practice

- ☐ Customer data never leaves agreed systems. Not to your laptop for convenience, not into a personal cloud drive, not into a third-party AI tool that is not covered by the agreement.
- ☐ Never name a customer to another customer without written permission, including in casual conversation and including in interviews.
- ☐ Anonymise examples properly before reuse. "A European steel distributor with nine CS agents" is not anonymous to anyone in that industry.
- ☐ Personal information told to you in confidence — a resignation, an illness, a dispute — is never repeated, never used as leverage, and never appears in a note.
- ☐ Screen-share one window, never your desktop. Close your stakeholder register, your other customers' folders and your email first, every time.
- ☐ Assume anything you write in their systems may be read by anyone in their organisation, including the person it is about.
- ☐ When you leave, delete what you agreed to delete, and confirm in writing that you have.

Conflicts of interest

TEXT

```
DECLARE, EARLY AND IN WRITING, IF:
 - You are working with a direct competitor of this customer
 - You have a financial interest in a vendor you are recommending
 - You previously worked for them, or for a party to a dispute
 - A family member or close friend is a stakeholder or supplier
 - Your firm's partnership with a vendor affects what you can recommend
 - You are recommending a tool your employer resells

THE DISCLOSURE FORMAT - simple, factual, unprompted:
 "Before we go further on the tooling decision: my employer is a
  certified partner for Vendor X and earns margin on their licences.
  That doesn't change my technical view, but you should know it and
  weight my recommendation accordingly. If you'd prefer an assessment
  from someone without that interest, I'll help you scope it."

WHAT NOT TO DO
 Wait to see whether it becomes relevant. It always becomes relevant,
 usually at the worst moment, and an undisclosed interest discovered by
 the customer is indistinguishable from concealment regardless of your
 actual intent.
```

When to tell a customer not to buy

TEXT

```
SAY NO WHEN:
 - The value case does not work and you cannot honestly make it work
 - The real blocker is organisational and software will not fix it
 - They need a different kind of help (a process change, a conversation
   between two directors, a decision somebody is avoiding)
 - You cannot staff it properly, and an under-resourced engagement will
   fail with your name on it
 - They are buying to avoid a harder decision
 - The data required does not exist and cannot be created economically
 - Another vendor genuinely fits better

HOW TO SAY IT
 "I don't think you should spend money on this, and here's why.

  The 26-hour response time is real. But three of the four causes are
  things you can change without any software: coverage hours, who owns
  the shared inbox, and whether supervisors can approve credit notes.
  Those are decisions, not systems.

  If you fix those first and the number is still bad, there is a
  genuine engineering problem underneath and I'd be glad to come back.
  But I'd be taking your money to build around a decision you can
  make for free, and that's not a good trade for you."

WHAT HAPPENS NEXT
 In roughly half of these conversations the customer buys something
 anyway, later, with a clearer problem and a bigger budget. In the
 other half you have lost revenue and gained a person who will take
 your call for the rest of your career. Both are acceptable outcomes.
 Pretending the project is sound is not.
```

Boundaries that protect you and them

Situation	Boundary	How to hold it
Asked to work the weekend "just this once"	Occasionally yes, as an exception, explicitly named as one	"I'll do this Saturday. I want to flag it's not sustainable — if the plan needs weekends, the plan is wrong."
Asked to do something outside the agreement	Small favours yes; new workstreams no	Time-box it and report the time in status, so it is visible
Asked to present numbers you do not believe	Hard no	"I can't present that number. Here's one I can defend."
Asked to omit a risk from a board paper	Hard no	"If it's not in the paper I can't be the author. I'm happy to word it carefully."
Drawn into internal politics	Stay out, visibly and symmetrically	"I genuinely don't have a view on that. What I can tell you is what the data shows."
Asked to evaluate a colleague's performance	Decline, redirect	"That's not something I can comment on. I can tell you what I observed about the process."
Offered a job by the customer	Disclose to your employer immediately	Most MSAs have a non-solicitation clause; silence creates a real problem
You find something seriously wrong (fraud, safety, legal)	Escalate through the agreed path, document, do not investigate alone	Tell your own firm's leadership the same day

The long game

Your reputation is a single asset carried across every engagement, every employer and every customer in an industry that is smaller than it looks. The people in the room with you today will be procurement directors, CTOs and founders in ten years, and they will remember exactly one thing: whether your assessment could be trusted when it was inconvenient. Every short-term gain from shading the truth is borrowed against that, at a punitive rate.

Interview Questions

Forward-deployed and solutions-engineering interviews probe judgement under ambiguity far more than algorithms. The strongest answers are specific, show a structure, include a number, admit a mistake, and end with what you would do differently. Vague process descriptions fail; worked examples pass.

A customer tells you in the kickoff meeting exactly what they want built. What do you do?

I take the request seriously as a signal, but I treat it as a hypothesis about the solution rather than a statement of the problem, because what a customer asks for is usually the solution they could imagine with the information they had. So I would say "that's helpful, and before I commit engineering time I want to understand the problem well enough to confirm that's the best use of the budget" — which almost nobody objects to, because it is obviously in their interest. Then I run a short discovery: interview the people who do the work rather than only the people who manage it, get a measured baseline from their own systems, and build an issue tree so I can size each branch. In one case a customer asked for a chatbot to speed up customer-service email; two weeks of measurement showed 19 of the 26-hour response time was queueing before an agent ever opened the mail, which a chatbot does not touch. I did not refuse to build it — I showed the arithmetic and let the sponsor choose, and they chose routing and assignment rules, which cost a fifth as much. The key is that I never said no; I made the cost of each option visible and handed the decision back. If discovery confirms the original request, that is also a good outcome: it costs two weeks and buys a confident, defensible scope.

How do you handle a stakeholder who is actively obstructing your project?

I start from the assumption that they have a rational interest I have not understood yet, because people are rarely obstructive for sport. So the first move is diagnostic: what do they lose if I succeed, have they been burned by something similar before, are they objecting to the substance or to the fact that they were not consulted, and — the one people skip — are they actually right? On one engagement the automation CoE lead kept reframing our work as duplicating his programme. The real interest was that his programme needed visible wins and ours would land first. Competing on technical merit would have been irrelevant to that, so I gave away the credit and kept the delivery: our rule shipped as a reusable component in his catalogue, with his team's branding, and he presented it at steering. It cost me nothing I needed and converted a blocker into a sponsor. The two things I avoid are escalating silently over someone — which permanently ends cooperation — and winning an argument publicly, because public defeats harden into permanent positions. The exception is when someone is factually wrong in a way that will cause real harm; then I say it once, calmly, in writing, and if I am overruled I record that it was overruled and by whom.

Your project is going to be three weeks late. Walk me through what you do.

I tell the sponsor the day I know, by phone, before anyone else hears it and before I have a complete recovery plan, because the decay rate on bad news is brutal — the same fact a week later reads as

concealment. The structure I use is fact, impact, options, recommendation, ask. Fact: the security review slot moved from the 25th to the 11th, which is a two-week shift on a gate everything else sits behind, and I should have submitted a week earlier than I did. Impact: stated precisely, because unquantified bad news gets catastrophised — three weeks, no cost impact, go-live moves from 28 March to 11 April. Options: full scope on 11 April; or a reduced go-live on 1 April with the two highest-value components and the rest following; or hold the original date by cutting the composite view entirely, decided this week. Recommendation: the second, because it protects the four-week measurement window they need before a September renewal conversation. Ask: what do you want to do, and who else needs to hear this from you before Thursday's steering, so my sponsor is never surprised in public. Then I hit the new date, because a missed recovery date costs far more than the original slip. I would also check whether anything in the SOW's assumptions section covers this, because a documented assumption failure turns an apology into a contractual formality.

How do you estimate work in an environment you have never seen before?

I decompose into items of half a day to three days, because anything estimated as "two weeks" is an un-decomposed lump concealing unknowns — and if I genuinely cannot break something down, I treat it as a research task and estimate the research instead of the build. Each item gets a three-point estimate and I pay more attention to the spread than the midpoint: the items where pessimistic is four times optimistic are where the budget actually gets consumed, and in enterprises those are almost always data access, security review and anything that depends on another team's queue. Then I explicitly estimate the non-code work — meetings, documentation, demos, chasing test data, training — which is 30 to 50 per cent of effort on an enterprise engagement and which engineers routinely cost at zero. Finally I apply a new-customer multiplier, typically 1.35 to 2.0 on a first engagement, which is not padding but the measured cost of working inside an organisation whose access processes and change windows I do not yet know. I never give a number verbally in a meeting; "I'll come back to you tomorrow" is always correct, because numbers given under social pressure become commitments. And I record estimate versus actual per item after each milestone, because within three engagements that gives me my own personal multiplier, which is worth more than any framework.

A customer asks for something you know is a bad idea, and they are senior. How do you respond?

I never contradict a senior person in public, and I never agree to something I will later have to unpick, so the move is to convert an opinion dispute into a measurable question and buy forty-eight hours. First I reframe to intent: "let me make sure I have the goal — you want fewer customers needing to contact us at all, is that right?" That separates the objective, which is usually sound, from the solution, which may not be. Then I offer one number if I have one — "of the 26 hours a customer waits, 19 is queueing and four is agent handling; a chatbot works on the four" — and propose coming back with two sized options and the expected effect of each, so the decision rests on numbers rather than on my opinion versus theirs. I always include their option in that comparison, properly costed and honestly argued, and I say explicitly that if its numbers are better I will build it. That gives a senior person a way to change their mind without losing face: they are not retreating from a position, they

are choosing between two options I brought them. Occasionally the analysis shows they were right, which is a good outcome for everyone and does a lot for my credibility the next time I disagree.

How do you quantify the business value of something that is hard to measure?

I separate what I can defend from what I cannot, and I refuse to invent numbers for the second category, because a value case that gets dismantled in public by a finance analyst destroys your credibility for every future number you present. Concretely: I build the hard case conservatively — hours returned at the company's own fully-loaded rate, run costs subtracted, adoption ramped rather than assumed at 100 per cent from day one — and I present the soft benefits as counted facts without a currency attached. On one engagement the hard saving was €23,400 a year against a €101,000 investment, which is a 5.7-year payback and does not pass any reasonable hurdle. I said that out loud before finance found it. Then I presented the case that actually justified the spend: 1,900 fewer delayed deliveries a year, elimination of a single-person dependency with no cover, and two key accounts worth €2.1 million in renewal value that had specifically cited responsiveness. I told the COO I could not price those but she could. I also always baseline before changing anything, using a metric that already exists in their systems, with the query written down and signed off by the person who will judge the result — because measuring "the same way" six months later is impossible otherwise, and an unbaselined success is an argument waiting to happen.

Tell me about a time you got a requirement badly wrong.

On an order-management engagement I built an address-matching rule from the documented process and a workshop, and it was wrong in two ways that only surfaced when the domain expert reviewed it in week six: Austrian postcodes have a different structure that broke the normalisation, and "c/o" lines were being treated as part of the street address, which produced false mismatches on about 4 per cent of cases. The root cause was mine — I had interviewed four managers and no practitioners, so I knew the process as documented, which is to say I knew fiction. What I changed, immediately and permanently, was that I now shadow before I design: I sit next to someone doing the actual work for at least a full cycle, I log every application switch and every pause, and I compare observed timings against reported ones, because the gap between them is usually the most valuable number of the week. I also made the SME the formal reviewer of every rule before anyone else saw it, framed as "I need you to break this" — which gave him the role he wanted, surfaced the remaining errors early, and turned a sceptic into the person who co-presented the solution to his own team. The cost of that mistake was about four days of rework. Caught in production it would have been a credibility event.

How do you manage scope creep without damaging the relationship?

I make the cost of every addition visible at the moment it is requested and then hand the decision back, because the damage comes from unilateral refusal, not from honest trade-offs. The mechanism is a change request with three options, an assessment and a recommendation — never a single option, which reads as a bill. When a sponsor asked to add the Austrian subsidiary mid-phase, the options were: include it fully for six extra days and an eleven-day slip; defer it to phase 2 as planned; or build only the reason-code mapping now for two days with no schedule impact, which turns phase

2 for Austria from a six-day project into a three-day deployment. I recommended the third, and finding that cheaper middle path is the part that makes you valuable rather than obstructive. The other half of the defence is the out-of-scope section of the SOW, which I write deliberately long — every line there is an argument I will not have in month five — and when a customer asks why it is so long I tell them honestly: so neither of us is surprised, and if any of it should be in scope, now is the cheapest moment to add it. I also log the small asks and show the running total in status, because creep is rarely one dramatic demand; it is forty half-day favours.

You have been dropped into a customer site with a vague brief and no documentation. What do you do in week one?

Week one is about three things: orientation, a measured baseline and visible output. On day one I find out who the economic buyer is — not the person who booked my flight, the person whose budget pays for this next year — and I ask my main contact directly, because if that answer is vague I am in a pilot with no path to production and I need to know that in week one rather than month five. I spend the first days interviewing across layers, deliberately including practitioners rather than only managers, and I shadow at least one person doing the real work for a full cycle. In parallel I start the long-lead items immediately: data access requests, VPN, security review submission, and anything that goes through someone else's ticket queue, because those are almost always the critical path and they do not care how urgent my project is. By the end of the week I write a one-page day-one answer — a straw man of what I think the problem is, what I think the solution is, and what we would have to believe for that to be true, with a test for each belief — and I present it labelled as a straw man and ask people to attack it. I also try to ship one small, visible, useful thing nobody asked for, like fixing a broken report, because demonstrated competence in week one buys more cooperation than any amount of credentials.

How do you get a decision out of an organisation that will not decide?

Three levers, in order. First, find out who actually decides, by asking the question literally — "who makes this call?" — because a surprising amount of enterprise paralysis is simply that nobody knows, and the question has never been asked out loud. Second, shrink the decision until it is reversible, because most deferral is risk aversion about irreversibility rather than disagreement about substance: "approve the programme" is a career risk, "approve a four-week pilot with nine users that we can switch off in ten minutes" is not, and once the small thing is working the large decision gets made retroactively by ratifying reality. Third, use default decisions with a deadline, agreed as a working practice up front: a five-line email with two options, a recommendation, and "unless you tell me otherwise by Friday I will proceed with A and record it as a default decision." The default must always be the safer and more reversible option, or the mechanism is manipulation rather than momentum. Underneath all of that is pre-wiring: no significant decision should be made for the first time in the meeting where it is formally taken. I spend about two hours before a steering committee walking the recommendation past the champion, the likely objector and the technical authority, asking each "what would make you say no to this?" — which surfaces in private the objection that would otherwise be raised in public.

What makes a good handover, and how do you know it worked?

> The test of a handover is not whether a document exists; it is whether a named person who was not on the project can perform real scenarios using only the written material. So I run it as an observed test: I pick three realistic failure modes, I sit in the room while their operator works through the runbooks, and I say nothing. Everywhere they hesitate is a gap, and I fix those and repeat before I call the handover accepted — that acceptance criterion goes in the SOW, phrased as their person succeeding rather than my document existing. The pack itself covers architecture, environments, a credentials inventory with what breaks when a named person leaves, runbooks per failure mode, monitoring and who alerts route to, known issues and accepted limitations, the deferred-debt register with estimates, data lineage and PII locations, the support model with hours and escalation, and the decision log. I also plan adoption from week two rather than week twenty, because a system nobody uses is a handover failure regardless of documentation: pilot with volunteers rather than big-bang, have the customer's own respected person deliver the second training session while I sit at the back silently, and watch for the shadow spreadsheet that reveals an unhandled edge case nobody wanted to mention. The best ending is their team demoing the system to another part of the business, correctly, without me present.

> **When would you tell a customer not to buy from you?**
>
> When the value case does not work and I cannot honestly make it work; when the real blocker is organisational and software will not fix it; when they are buying software to avoid a decision somebody is reluctant to make; when I cannot staff it properly, because an under-resourced engagement fails with my name on it; or when another vendor genuinely fits better. The way I say it is factual rather than noble: "I don't think you should spend money on this yet. Three of the four causes of your 26-hour response time are decisions — coverage hours, who owns the shared inbox, and whether supervisors can approve credit notes — and those cost nothing. Fix them, re-measure, and if there's still an engineering problem underneath I'd be glad to come back." Roughly half those customers buy something later with a clearer problem and a bigger budget; the other half become people who take my call for the rest of my career. Both outcomes beat taking money to build around a free decision, which produces a project that cannot succeed and a reference I would not want. The same principle governs declaring conflicts of interest unprompted — if my employer earns margin on a product I am recommending, the customer hears that from me before the recommendation, not from someone else afterwards.

Drills and a 30-Day Practice Plan

Consulting skills are motor skills. Reading about laddering does nothing; asking "what does that mean concretely?" fifty times in real conversations changes how you listen permanently. The drills below are designed to be run inside whatever work you already have — no customer engagement required — and each produces an artefact you can review.

Daily drills (10–20 minutes each)

Drill	What you do	Artefact
The summary email	After every meeting, send: what I heard, what we decided, what I'll do, what I need, by when	One email, within two hours
So what?	Take one fact from your day and push it through three "so what?" steps to a business consequence	Three lines in a notebook
Ladder down	In one conversation, chase one abstraction to a concrete fact: "what does 'slow' mean in seconds?"	A number you did not have
The commitment ledger	Write down every promise you made today, including corridor ones	A list you clear each morning
One verbatim quote	Capture one thing someone said, in their exact words	A growing quote bank
Five seconds	After someone finishes speaking, count five before replying, once per day	Note what they added in the gap

Weekly drills (45–90 minutes)

1. **Build an issue tree**
   for any problem in your own work. Three levels, MECE, size every branch even if you guess, strike through the branches you are not pursuing. Then show it to someone and ask where it is wrong.

2. **Write a problem statement canvas**
   for something you are currently building. All seven fields. The fields you cannot fill are your week's research list.

3. **Run one real discovery interview**
   with a colleague about a process you do not understand. Use the question bank. Write the full write-up template, including the "my inferences" section, and send it to them for correction.

4. **Shadow someone for an hour**
   using the observation log format. Compare their reported task time to your measured time and note the ratio.

5. **Draft a weekly status email**

for your current work even if nobody asked for one. Include a RAG with an explicit trigger and exit condition, and one decision request above the fold.

6 **Write a three-option change request**
for the last scope change you absorbed silently.

7 **Pre-wire one decision.**
Before your next team decision, speak to three people individually and ask "what would make you say no to this?"

The 30-day plan

TEXT

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WEEK 1 — LISTENING
 Goal: stop proposing solutions in the first ten minutes of a conversation.
 Mon  Read back the interview question bank; pick 10 you will actually use.
 Tue  Run one discovery interview (45 min) with a colleague. Record your
      question/talk ratio afterwards — target: they talk 70%+.
 Wed  Write the full interview write-up. Send it for correction.
 Thu  Shadow someone for one hour. Observation log. Note reported vs measured.
 Fri  Review the week: how many times did you propose a solution before
      you had a number? Write that count down. It is your baseline.
 DAILY: summary email after every meeting. No exceptions.
 ARTEFACTS: 1 interview note, 1 observation log, 5 summary emails.

WEEK 2 — STRUCTURING
 Goal: never again present an unstructured finding.
 Mon  Issue tree for your current project's main problem. 3 levels, sized.
 Tue  Write the day-one answer: situation, complication, answer, what we
      must believe, kill criteria. One page. Show it to your lead as a
      straw man and ask them to attack it.
 Wed  Problem statement canvas, all 7 fields, for the same problem.
 Thu  Pareto something real. Pull actual data. Count on the unit that
      matters, not the unit that is easy.
 Fri  Write a hypothesis card for your biggest open assumption, with a
      falsifying number and a test that costs under four hours. Run it.
 ARTEFACTS: issue tree, day-one answer, canvas, one Pareto, one tested
 hypothesis — including, ideally, one you disproved.

WEEK 3 — STAKEHOLDERS AND COMMUNICATION
 Goal: know who decides, and never surprise anyone.
 Mon  Stakeholder register for your current work. Name the economic buyer.
      If you cannot, that is this week's most important finding.
 Tue  Power/interest map. Identify your champion, your most likely blocker
      and your gatekeeper by name.
 Wed  Engagement plan: who hears what, from whom, how often. One page.
 Thu  Write and send a full weekly status email with a justified RAG, a
      decision request above the fold, and an honest burn figure.
 Fri  Pre-wire one decision: three private conversations asking "what
      would make you say no to this?" before the meeting where it is taken.
 ARTEFACTS: register, map, engagement plan, 1 status email, 1 pre-wired
 decision that took under fifteen minutes to agree.

WEEK 4 — SCOPE, VALUE AND CLOSING
 Goal: be able to defend a number and a boundary.
 Mon  Write an out-of-scope list for your current work. Aim for 10 items.
      Circulate it and watch which ones cause a reaction — those are the
      ones that would have become arguments.
 Tue  Build the value model: 5 tabs, all assumptions on one tab with
      source and confidence, a sensitivity table on the two weakest inputs.
 Wed  Write the acceptance criteria for your current deliverables such
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     that someone who was not in the room could test them.
Thu  Draft the handover pack table of contents. Answer honestly: "if I
     were unreachable for six weeks starting tomorrow, what breaks?"
Fri  Script one hard conversation you have been avoiding, using fact /
     impact / options / recommendation / ask. Then have it.
ARTEFACTS: out-of-scope list, value model, acceptance criteria,
handover TOC, one difficult conversation actually completed.
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Self-assessment at day 30

- ☐ I can name the economic buyer of my current work and the last thing they said to me directly.
- ☐ I have a written, agreed problem statement with a number in it and a stated source.
- ☐ I have an issue tree where every branch is sized and the pruned branches are visible.
- ☐ Every meeting I attended this month was followed by a summary email within two hours.
- ☐ I have told someone bad news early at least once, and it was a non-event.
- ☐ I have a written out-of-scope list that someone else has read.
- ☐ I have disproved at least one of my own hypotheses in public.
- ☐ I have asked "what should I have asked you that I didn't?" and got a useful answer.
- ☐ My last status report included an amber or a miss, with a specific exit condition.
- ☐ I can state what breaks if I am unreachable for six weeks, and I am fixing it.

✓ **The compounding skill**

Of everything in this guide, the habit with the highest return per unit of effort is the post-meeting summary email: what I heard, what we decided, what I will do, what I need from you, by when. It takes six minutes. It creates the written record, surfaces misunderstandings while they are still cheap, makes you the author of the agreed version of events, and — because almost nobody does it consistently — marks you within a month as the person in the room who can be relied upon. Start there, today, and let the rest follow.